Análisis exploratorio dataset - Airbnb

Introducción

In [43]:
```
import numpy  as np
import pandas as pd

import matplotlib.pyplot as plt
%matplotlib inline

import warnings
warnings.filterwarnings('ignore')
```

1. Carga de datos y división train/test

Para esta exploración de datos, se decide crear dos archivos train/test para la exploración de los datos y construcción del modelo, esto nos permite evitar el data leakage. Adicional se define que lo datos van a estar separados en 70% Train y 30% Test.

In [44]:
```
# Importamos la librería para dividir el dataset en train y test
from sklearn.model_selection import train_test_split

# Cargamos el dataset completo en un DataFrame de pandas "full_df"
full_df = pd.read_csv("./airbnb-listings-extract.csv", delimiter=';')
train, test = train_test_split(full_df, test_size=0.3, shuffle=True, random_state=0

# Dimensiones de los datasets de training y test
print('Dimensiones de los datasets:')
print(f'Original 100% de los datos: {full_df.shape}')
print(f'Training 70% de los datos: {train.shape}')
print(f'Test 30% de los datos: {test.shape}')

# Se guardan los datasets de training y test en formato CSV, con el separador ';' y
train.to_csv('./airbnb_train.csv', sep=';', decimal='.', index=False)
test.to_csv('./airbnb_test.csv', sep=';', decimal='.', index=False)
```

```
Dimensiones de los datasets:
Original 100% de los datos: (14780, 89)
Training 70% de los datos: (10346, 89)
Test 30% de los datos: (4434, 89)
```

2. Análisis exploratorio

In [45]:
```
# Cargamos el dataset de training en un DataFrame de pandas "airbnb_df"
airbnb_df_full = pd.read_csv('./airbnb_train.csv', sep=';', decimal='.')

# Mostramos las primeras 5 filas del dataset de training (la funcion .T nos permite
airbnb_df_full.head(5)
```

Out[45]:

	ID	Listing Url	Scrape ID	Last Scraped	Nan
0	2156319	https://www.airbnb.com/rooms/2156319	20170407214119	2017-04-08	GRANT Plaza May
1	3377153	https://www.airbnb.com/rooms/3377153	20170407214119	2017-04-08	Nice flat Plaza May (li
2	14800635	https://www.airbnb.com/rooms/14800635	20170407214119	2017-04-08	* ROO double Bar Salamanca
3	11185674	https://www.airbnb.com/rooms/11185674	20170407214119	2017-04-08	che apartment Madrid c ce
4	18092171	https://www.airbnb.com/rooms/18092171	20170407214119	2017-04-08	ACOGEDC APARTAMENT EN MADRI CAL MANZANAR

5 rows × 89 columns

In [46]:
```
#Se validan todas las columnas del Dataset, cantidad de nulos y tipo de dato.
airbnb_df_full.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10346 entries, 0 to 10345
Data columns (total 89 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   ID                              10346 non-null  int64
 1   Listing Url                     10346 non-null  object
 2   Scrape ID                       10346 non-null  int64
 3   Last Scraped                    10346 non-null  object
 4   Name                            10345 non-null  object
 5   Summary                         9922 non-null   object
 6   Space                           7619 non-null   object
 7   Description                     10340 non-null  object
 8   Experiences Offered             10346 non-null  object
 9   Neighborhood Overview           6397 non-null   object
 10  Notes                           3960 non-null   object
 11  Transit                         6327 non-null   object
 12  Access                          5814 non-null   object
 13  Interaction                     5747 non-null   object
 14  House Rules                     6738 non-null   object
 15  Thumbnail Url                   8319 non-null   object
 16  Medium Url                      8319 non-null   object
 17  Picture Url                     10329 non-null  object
 18  XL Picture Url                  8319 non-null   object
 19  Host ID                         10346 non-null  int64
 20  Host URL                        10346 non-null  object
 21  Host Name                       10344 non-null  object
 22  Host Since                      10344 non-null  object
 23  Host Location                   10318 non-null  object
 24  Host About                      6692 non-null   object
 25  Host Response Time              9016 non-null   object
 26  Host Response Rate              9016 non-null   float64
 27  Host Acceptance Rate            25 non-null     object
 28  Host Thumbnail Url              10344 non-null  object
 29  Host Picture Url                10344 non-null  object
 30  Host Neighbourhood              7665 non-null   object
 31  Host Listings Count             10344 non-null  float64
 32  Host Total Listings Count       10344 non-null  float64
 33  Host Verifications              10340 non-null  object
 34  Street                          10346 non-null  object
 35  Neighbourhood                   6706 non-null   object
 36  Neighbourhood Cleansed          10346 non-null  object
 37  Neighbourhood Group Cleansed    9645 non-null   object
 38  City                            10342 non-null  object
 39  State                           10240 non-null  object
 40  Zipcode                         9999 non-null   object
 41  Market                          10307 non-null  object
 42  Smart Location                  10346 non-null  object
 43  Country Code                    10346 non-null  object
 44  Country                         10345 non-null  object
 45  Latitude                        10346 non-null  float64
 46  Longitude                       10346 non-null  float64
 47  Property Type                   10346 non-null  object
 48  Room Type                       10346 non-null  object
 49  Accommodates                    10346 non-null  int64
 50  Bathrooms                       10308 non-null  float64
```

```
 51  Bedrooms                        10326 non-null  float64
 52  Beds                            10312 non-null  float64
 53  Bed Type                        10346 non-null  object
 54  Amenities                       10215 non-null  object
 55  Square Feet                     410 non-null    float64
 56  Price                           10332 non-null  float64
 57  Weekly Price                    2513 non-null   float64
 58  Monthly Price                   2502 non-null   float64
 59  Security Deposit                4447 non-null   float64
 60  Cleaning Fee                    6124 non-null   float64
 61  Guests Included                 10346 non-null  int64
 62  Extra People                    10346 non-null  int64
 63  Minimum Nights                  10346 non-null  int64
 64  Maximum Nights                  10346 non-null  int64
 65  Calendar Updated                10346 non-null  object
 66  Has Availability                7 non-null      object
 67  Availability 30                 10346 non-null  int64
 68  Availability 60                 10346 non-null  int64
 69  Availability 90                 10346 non-null  int64
 70  Availability 365                10346 non-null  int64
 71  Calendar last Scraped           10346 non-null  object
 72  Number of Reviews               10346 non-null  int64
 73  First Review                    8100 non-null   object
 74  Last Review                     8099 non-null   object
 75  Review Scores Rating            7995 non-null   float64
 76  Review Scores Accuracy          7976 non-null   float64
 77  Review Scores Cleanliness       7980 non-null   float64
 78  Review Scores Checkin           7971 non-null   float64
 79  Review Scores Communication     7980 non-null   float64
 80  Review Scores Location          7969 non-null   float64
 81  Review Scores Value             7967 non-null   float64
 82  License                         251 non-null    object
 83  Jurisdiction Names              143 non-null    object
 84  Cancellation Policy             10346 non-null  object
 85  Calculated host listings count  10342 non-null  float64
 86  Reviews per Month               8100 non-null   float64
 87  Geolocation                     10346 non-null  object
 88  Features                        10345 non-null  object
dtypes: float64(23), int64(13), object(53)
memory usage: 7.0+ MB
```

In [47]:

```
#Columnas consultadas para evaluar su contenido y determinar si se eliminan

airbnb_df_full[['ID','Neighbourhood','Neighbourhood Cleansed','Neighbourhood Group

#airbnb_df_full[['ID','Market']]
#airbnb_df_full['Market'].value_counts()

airbnb_df_full[['ID','Smart Location']]
airbnb_df_full['Smart Location'].value_counts()

#airbnb_df_full[['ID','Country Code']]

#airbnb_df_full[['ID','Property Type']]
#airbnb_df_full['Property Type'].value_counts()
```

```
#airbnb_df_full[['ID','Room Type']]

#airbnb_df_full[['ID','Accommodates']]
#airbnb_df_full['Accommodates'].value_counts()

#airbnb_df_full[['ID','Square Feet']]
#airbnb_df_full['Square Feet'].value_counts()

#airbnb_df_full[['ID','Calculated host listings count']]
#airbnb_df_full[['ID','Features']]

airbnb_df_full[['ID','Country']]
airbnb_df_full['Country'].value_counts()
```

Out[47]:

```
Country
Spain             9817
United Kingdom     118
United States      115
France              72
Italy               51
Australia           33
Germany             30
Ireland             23
Austria             22
Netherlands         16
Canada              14
Belgium             14
Denmark             10
Switzerland          4
Greece               3
Hong Kong            2
Cuba                 1
Name: count, dtype: int64
```

In [48]:

```
#Extraemos las columnas que consideramos mas relevantes por nombre de la columna
#(sentido común y validando el contenido) a un nuevo df

airbnb_df = airbnb_df_full[['ID', 'Street', 'Neighbourhood', 'Neighbourhood Cleanse
                            'Neighbourhood Group Cleansed', 'City', 'State', 'Smar
                            'Country Code', 'Country','Property Type','Room Type',
                            'Bathrooms','Bedrooms','Price','Number of Reviews','Fi
                            'Reviews per Month']]
```

In [49]:

```
#Exploramos los registros con valor null y se reemplzan manualmente de acuerdo a la
# información de otras columnas.

airbnb_df[airbnb_df['City'].isna()]
airbnb_df['City'].value_counts()
airbnb_df['City'].fillna('Madrid', inplace=True)
vcity_na = [1736, 1963, 4086, 4549]
airbnb_df.iloc[vcity_na]

airbnb_df[airbnb_df['Country'].isna()]
airbnb_df['Country'].value_counts()
airbnb_df['Country'].fillna('Spain', inplace=True)
```

```
airbnb_df[2231:2232]
airbnb_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10346 entries, 0 to 10345
Data columns (total 20 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   ID                            10346 non-null  int64
 1   Street                        10346 non-null  object
 2   Neighbourhood                 6706 non-null   object
 3   Neighbourhood Cleansed        10346 non-null  object
 4   Neighbourhood Group Cleansed  9645 non-null   object
 5   City                          10346 non-null  object
 6   State                         10240 non-null  object
 7   Smart Location                10346 non-null  object
 8   Country Code                  10346 non-null  object
 9   Country                       10346 non-null  object
 10  Property Type                 10346 non-null  object
 11  Room Type                     10346 non-null  object
 12  Accommodates                  10346 non-null  int64
 13  Bathrooms                     10308 non-null  float64
 14  Bedrooms                      10326 non-null  float64
 15  Price                         10332 non-null  float64
 16  Number of Reviews             10346 non-null  int64
 17  First Review                  8100 non-null   object
 18  Last Review                   8099 non-null   object
 19  Reviews per Month             8100 non-null   float64
dtypes: float64(4), int64(3), object(13)
memory usage: 1.6+ MB
```

In [50]:
```
#El 95% de los datos corresponden a España, por lo que se determina que el modelo s
# para España por que los datos de otros paises no son suficientes, solo nos quedam

airbnb_df = airbnb_df[airbnb_df['Country'] == 'Spain']
print(f'Nuevo dataset: {airbnb_df.shape}')
```

```
Nuevo dataset: (9818, 20)
```

In [51]:
```
#Validamos los campos numericos con nulos e imputamos los datos.
airbnb_df['Bathrooms'].value_counts()
airbnb_df['Bathrooms'].fillna(airbnb_df['Bathrooms'].mode()[0], inplace=True)

airbnb_df['Bedrooms'].value_counts()
airbnb_df['Bedrooms'].fillna(airbnb_df['Bedrooms'].mode()[0], inplace=True)

airbnb_df['Price'].fillna(airbnb_df['Price'].mode()[0], inplace=True)

airbnb_df.isnull().any()
```

Out[51]:
```
ID                              False
Street                          False
Neighbourhood                    True
Neighbourhood Cleansed          False
Neighbourhood Group Cleansed     True
City                            False
State                            True
Smart Location                  False
Country Code                    False
Country                         False
Property Type                   False
Room Type                       False
Accommodates                    False
Bathrooms                       False
Bedrooms                        False
Price                           False
Number of Reviews               False
First Review                     True
Last Review                      True
Reviews per Month                True
dtype: bool
```

In [52]:
```
# Se eliminan las columnas que no es posible imputar
airbnb_df = airbnb_df.drop(['ID','Neighbourhood','Neighbourhood Group Cleansed','St
airbnb_df.isnull().any()
```

Out[52]:
```
Street                    False
Neighbourhood Cleansed    False
City                      False
Smart Location            False
Property Type             False
Room Type                 False
Accommodates              False
Bathrooms                 False
Bedrooms                  False
Price                     False
Number of Reviews         False
dtype: bool
```

In [54]:
```
#Reemplazo de nombres y verificación de la información
airbnb_df.columns = ['Street', 'Neighbourhood', 'City', 'Smart_Location',
                     'Property_Type', 'Room_Type', 'Accommodates',
                     'Bathrooms', 'Bedrooms', 'Price', 'Total_Reviews']
print('datset: airbnb_df')
airbnb_df.info()
```

```
datset: airbnb_df
<class 'pandas.core.frame.DataFrame'>
Index: 9818 entries, 0 to 10345
Data columns (total 11 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Street          9818 non-null   object
 1   Neighbourhood   9818 non-null   object
 2   City            9818 non-null   object
 3   Smart_Location  9818 non-null   object
 4   Property_Type   9818 non-null   object
 5   Room_Type       9818 non-null   object
 6   Accommodates    9818 non-null   int64
 7   Bathrooms       9818 non-null   float64
 8   Bedrooms        9818 non-null   float64
 9   Price           9818 non-null   float64
 10  Total_Reviews   9818 non-null   int64
dtypes: float64(3), int64(2), object(6)
memory usage: 920.4+ KB
```

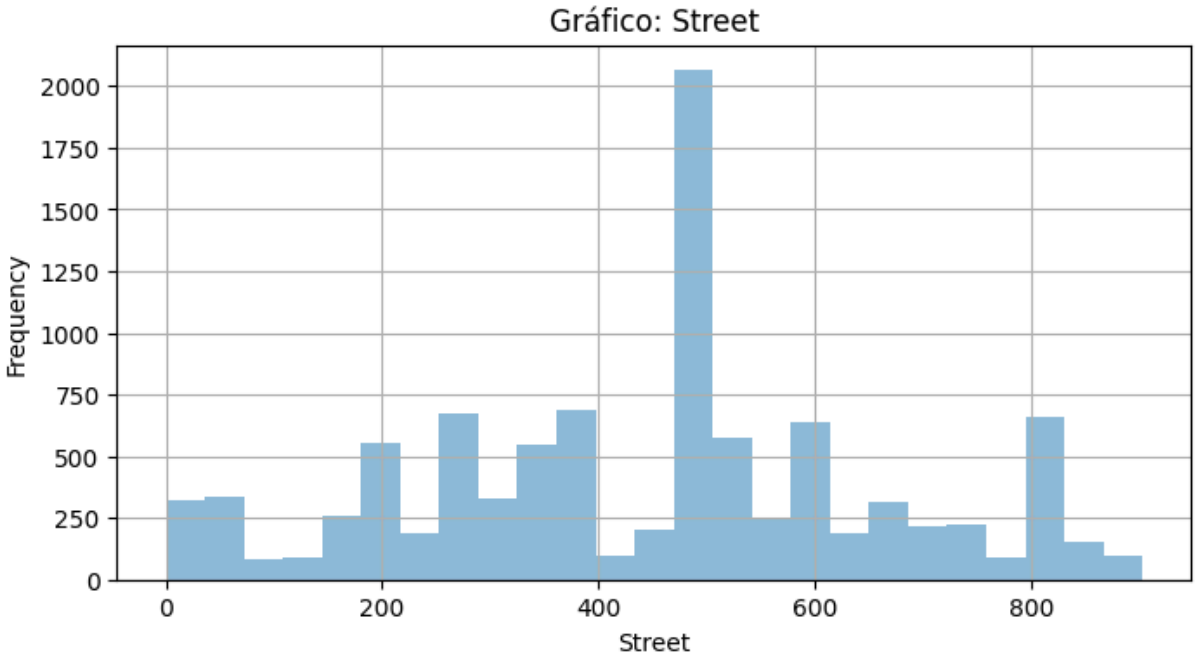
In [57]:

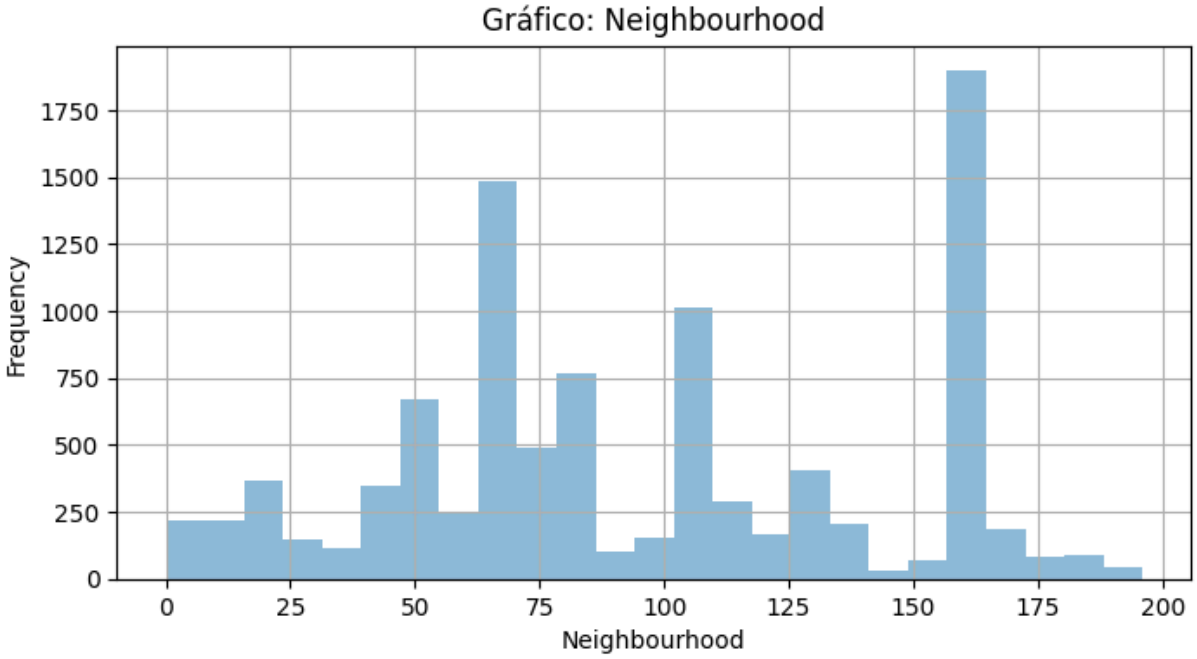
```
#Aplicamos el enconder para convertir todas las variables de tipo object
#El TargetEncoder se puede utilizar pero es necesario hacer la categorización con l
# que buscando un información genera Data Lakage y Overfitting, por lo que la opció
from sklearn.preprocessing import OrdinalEncoder

encoder = OrdinalEncoder()
airbnb_df[['Street']] = encoder.fit_transform(airbnb_df[['Street']])
airbnb_df[['Neighbourhood']] = encoder.fit_transform(airbnb_df[['Neighbourhood']])
airbnb_df[['City']] = encoder.fit_transform(airbnb_df[['City']])
airbnb_df[['Smart_Location']] = encoder.fit_transform(airbnb_df[['Smart_Location']]
airbnb_df[['Property_Type']] = encoder.fit_transform(airbnb_df[['Property_Type']])
airbnb_df[['Room_Type']] = encoder.fit_transform(airbnb_df[['Room_Type']])
airbnb_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 9818 entries, 0 to 10345
Data columns (total 11 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Street          9818 non-null   float64
 1   Neighbourhood   9818 non-null   float64
 2   City            9818 non-null   float64
 3   Smart_Location  9818 non-null   float64
 4   Property_Type   9818 non-null   float64
 5   Room_Type       9818 non-null   float64
 6   Accommodates    9818 non-null   int64
 7   Bathrooms       9818 non-null   float64
 8   Bedrooms        9818 non-null   float64
 9   Price           9818 non-null   float64
 10  Total_Reviews   9818 non-null   int64
dtypes: float64(9), int64(2)
memory usage: 920.4 KB
```

3. Visualización de variables

In [58]:
```
#Graficamos en un histograma
for c in airbnb_df.columns:
    plt.figure(figsize=(8, 4))
    airbnb_df[c].plot.hist(alpha=0.5, bins=25, grid = True)
    plt.title(f'Gráfico: {c}')
    plt.xlabel(c)

plt.show()
```





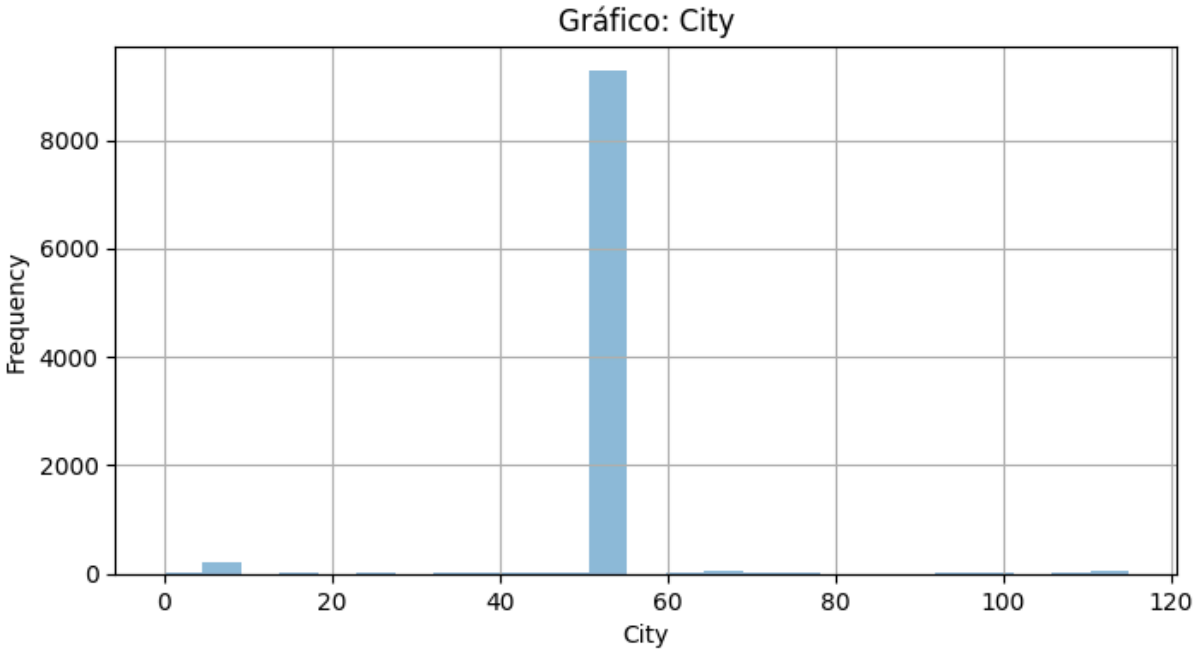


Gráfico: City
Frequency
8000
6000
4000
2000
0
0
20
40
60
80
100
120
City

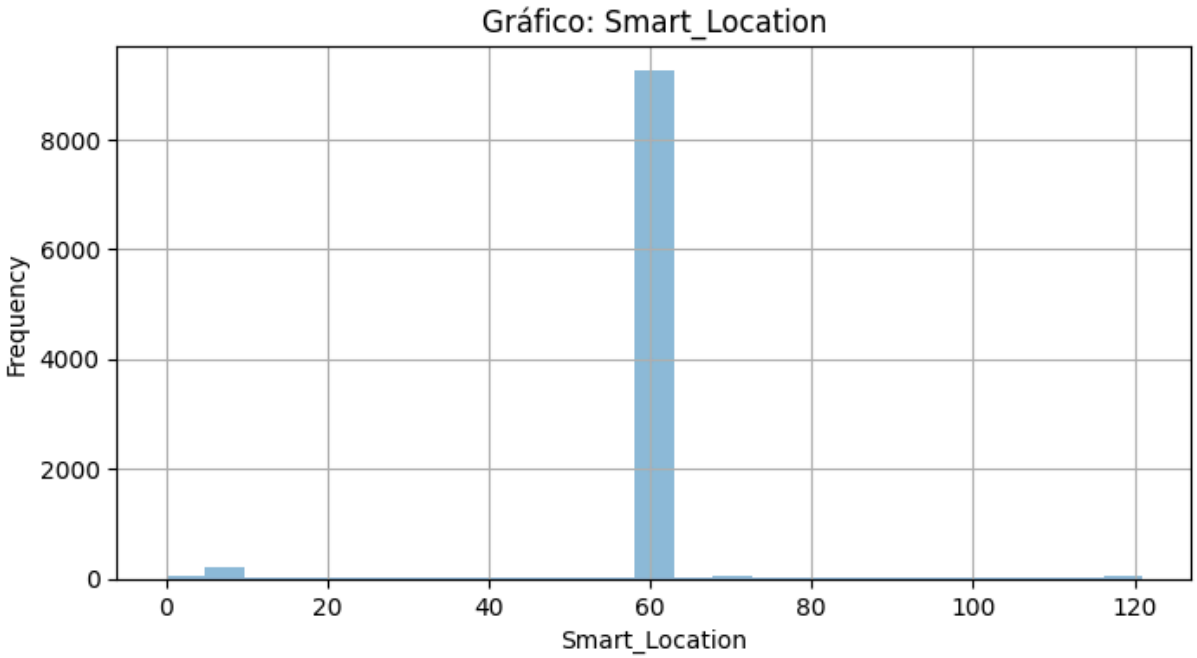


Gráfico: Smart_Location
Frequency
8000
6000
4000
2000
0
0
20
40
60
80
100
120
Smart_Location

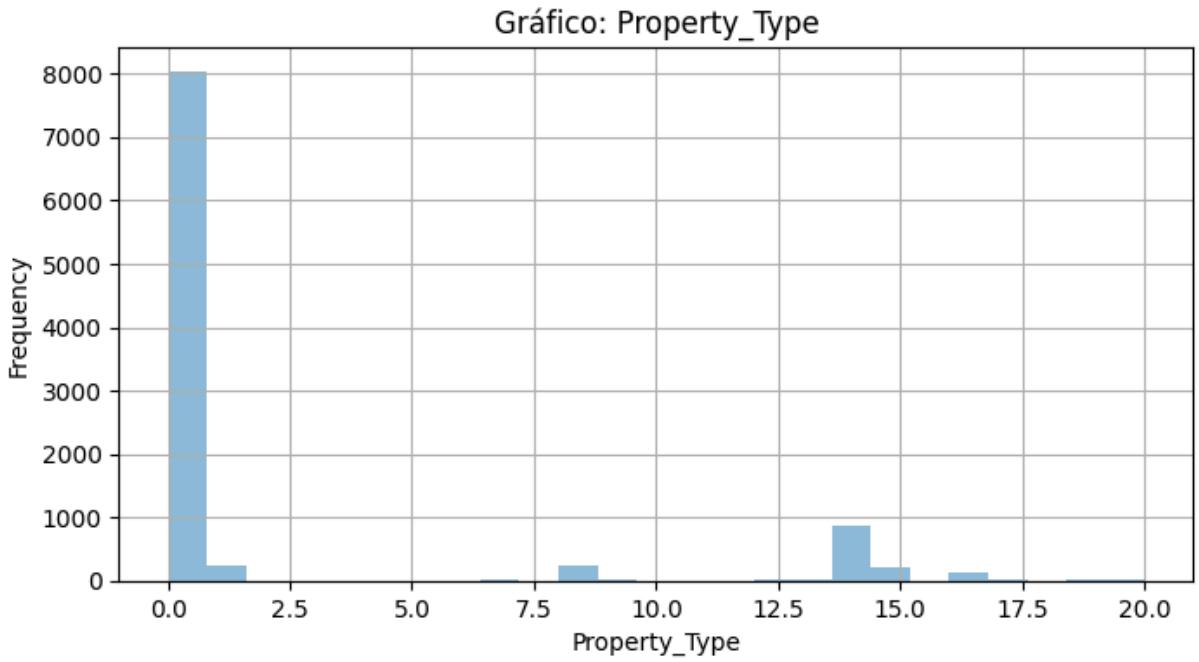


Gráfico: Property_Type
Frequency
8000
7000
6000
5000
4000
3000
2000
1000
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Property_Type

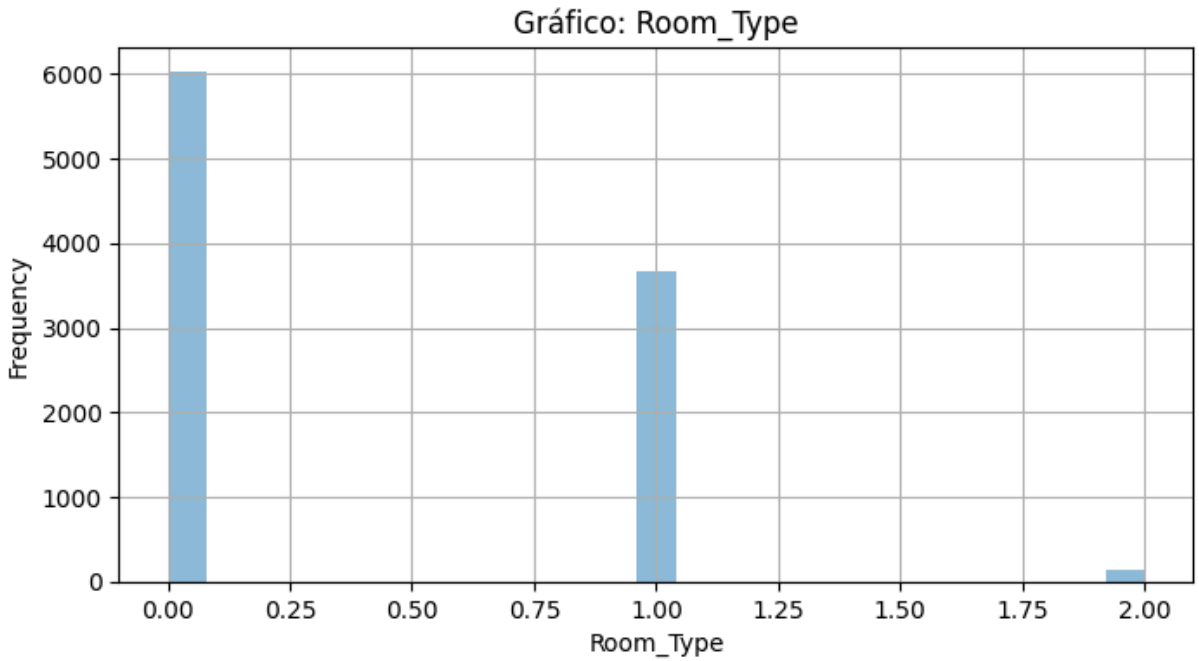


Gráfico: Room_Type
Frequency
6000
5000
4000
3000
2000
1000
0
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Room_Type

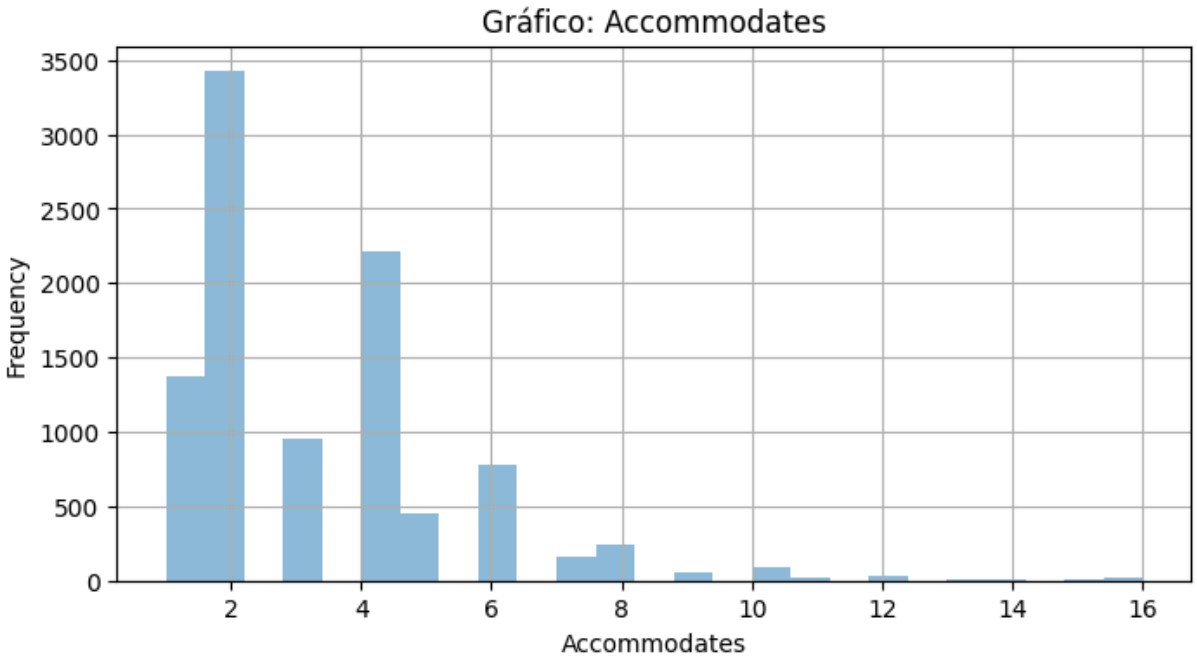


Gráfico: Accommodates
Frequency
Accommodates
0
500
1000
1500
2000
2500
3000
3500
2
4
6
8
10
12
14
16

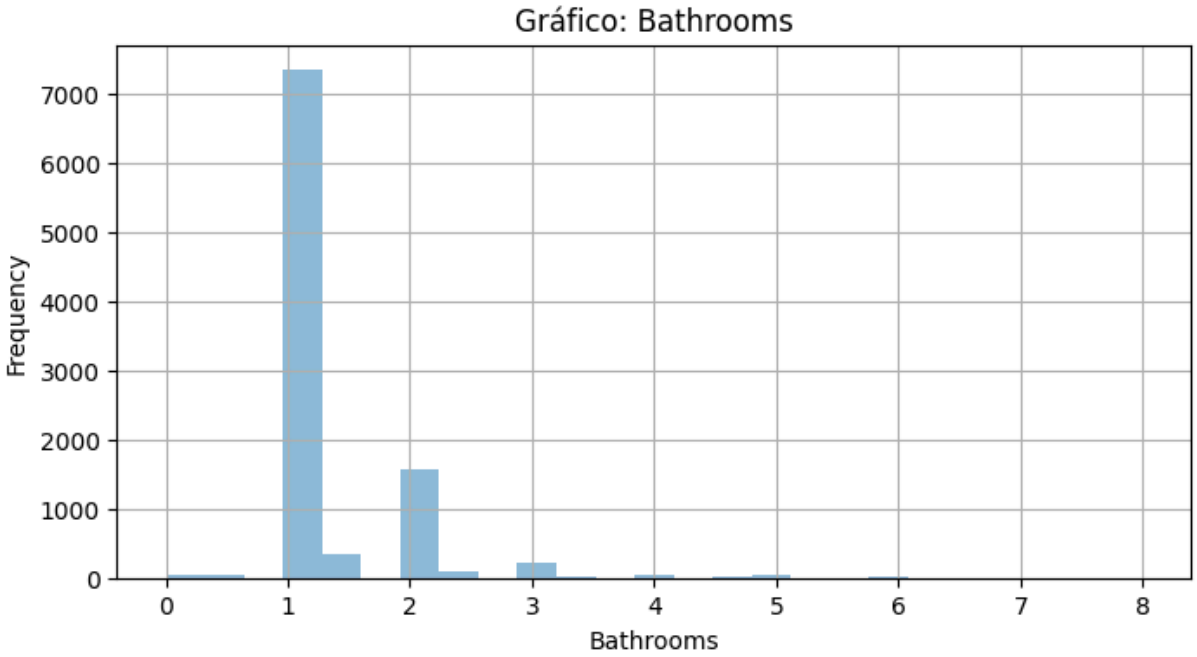


Gráfico: Bathrooms
Frequency
Bathrooms
0
1000
2000
3000
4000
5000
6000
7000
0
1
2
3
4
5
6
7
8

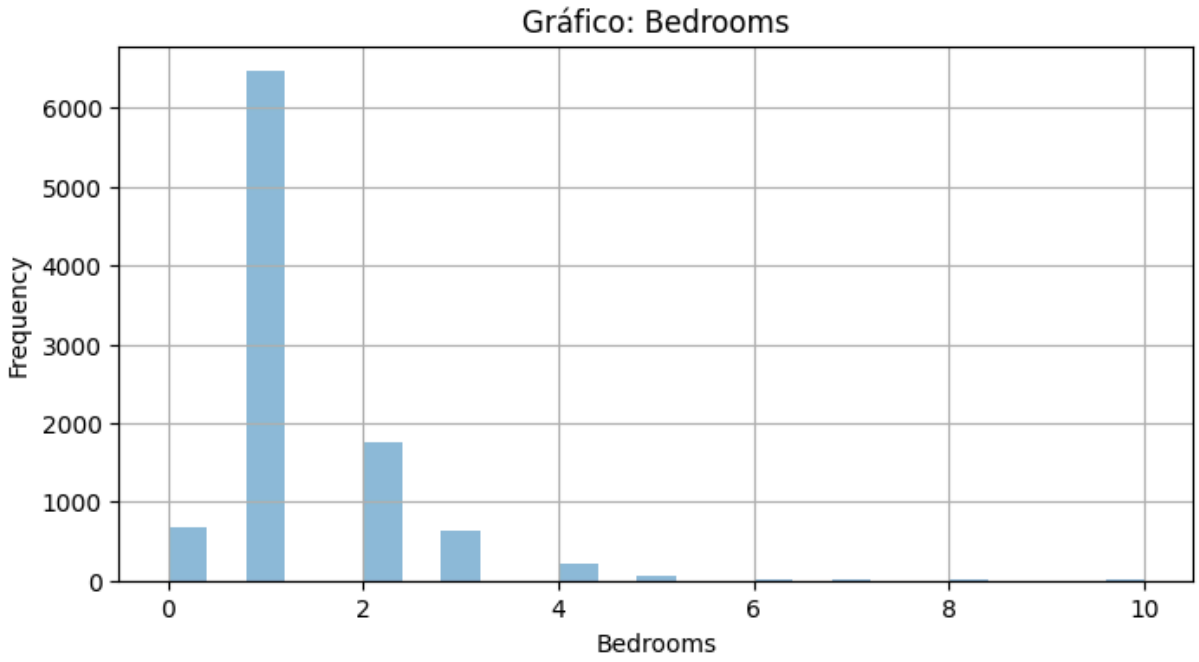


Gráfico: Bedrooms
Frequency
Bedrooms
0
1000
2000
3000
4000
5000
6000
0
2
4
6
8
10

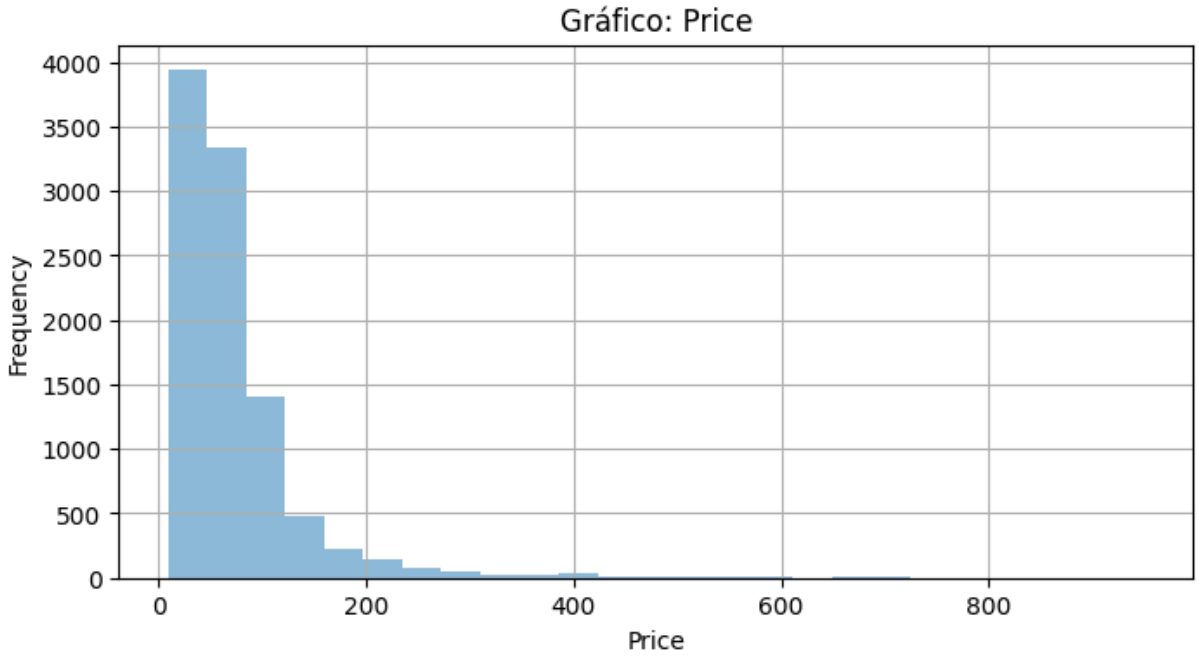
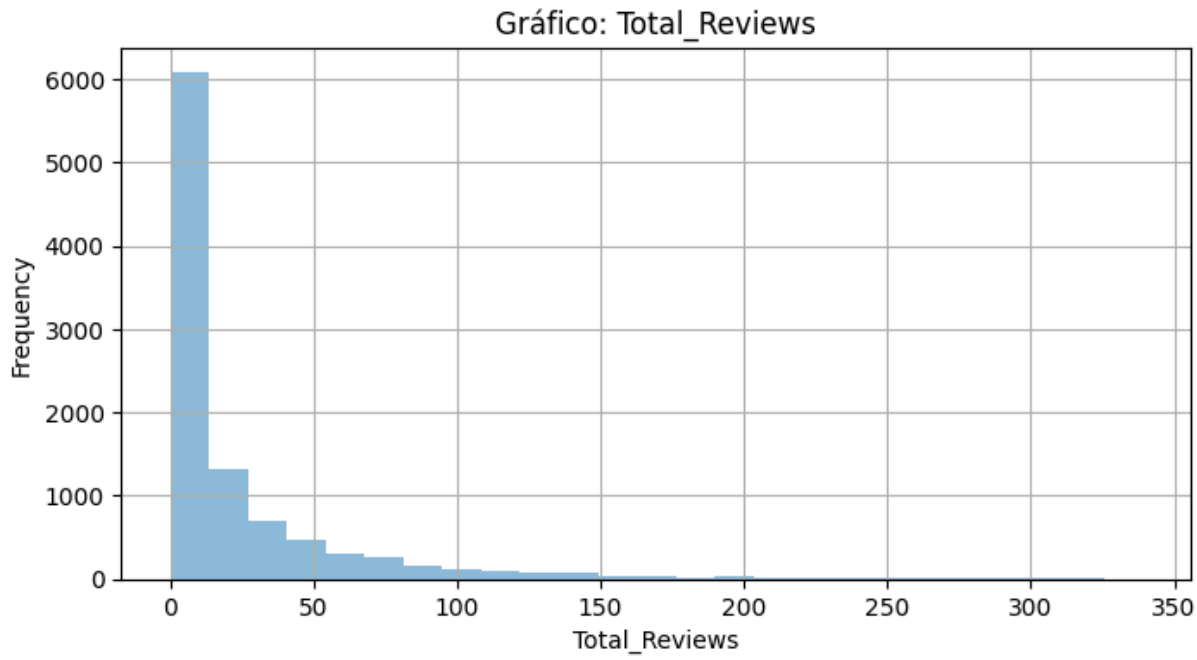
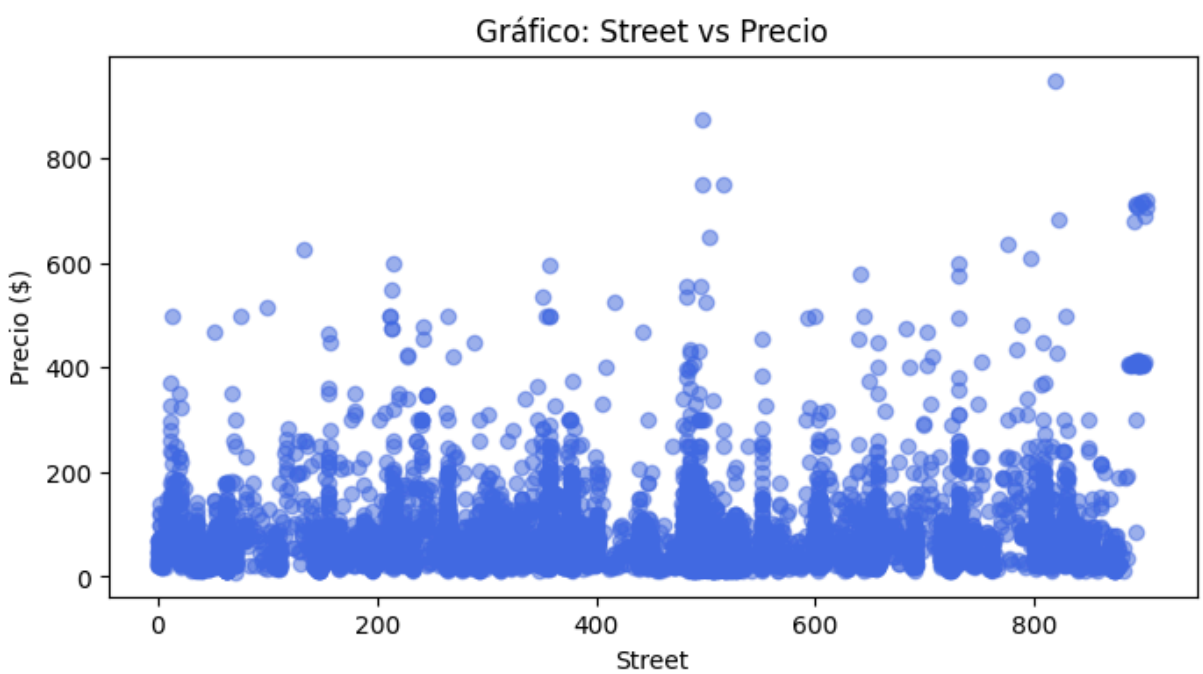


Gráfico: Price
Frequency
Price
0
500
1000
1500
2000
2500
3000
3500
4000
0
200
400
600
800



In [59]:
```
#Graficamos en un histograma
for c in airbnb_df.columns:
    plt.figure(figsize=(8, 4))
    plt.scatter(airbnb_df[c], airbnb_df['Price'], alpha=0.5, color='royalblue')
    plt.title(f'Gráfico: {c} vs Precio')
    plt.xlabel(c)
    plt.ylabel('Precio ($)')

plt.show()
```



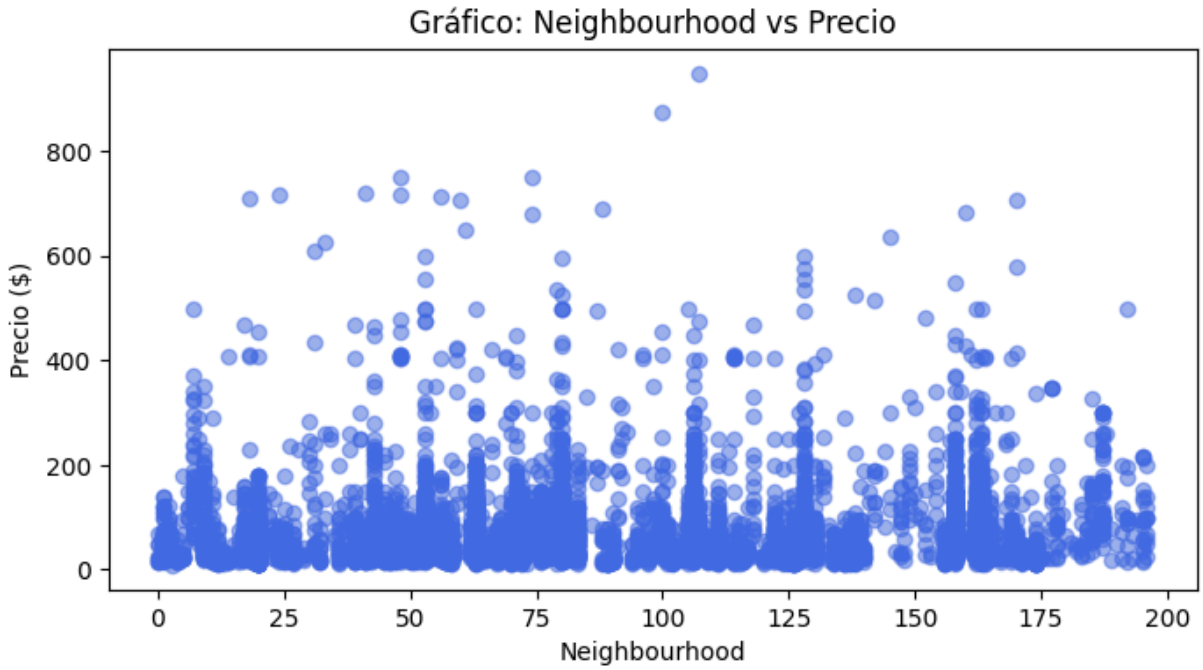


Gráfico: Neighbourhood vs Precio
Precio ($)
0
200
400
600
800
0
25
50
75
100
125
150
175
200
Neighbourhood

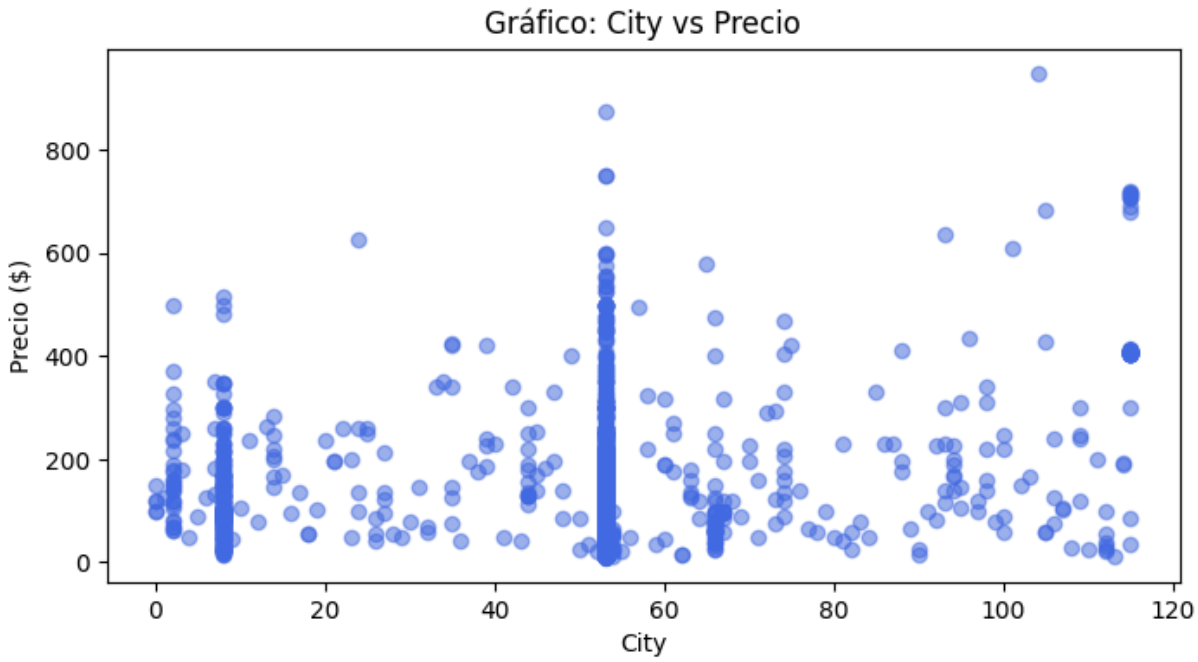


Gráfico: City vs Precio
Precio ($)
0
200
400
600
800
0
20
40
60
80
100
120
City

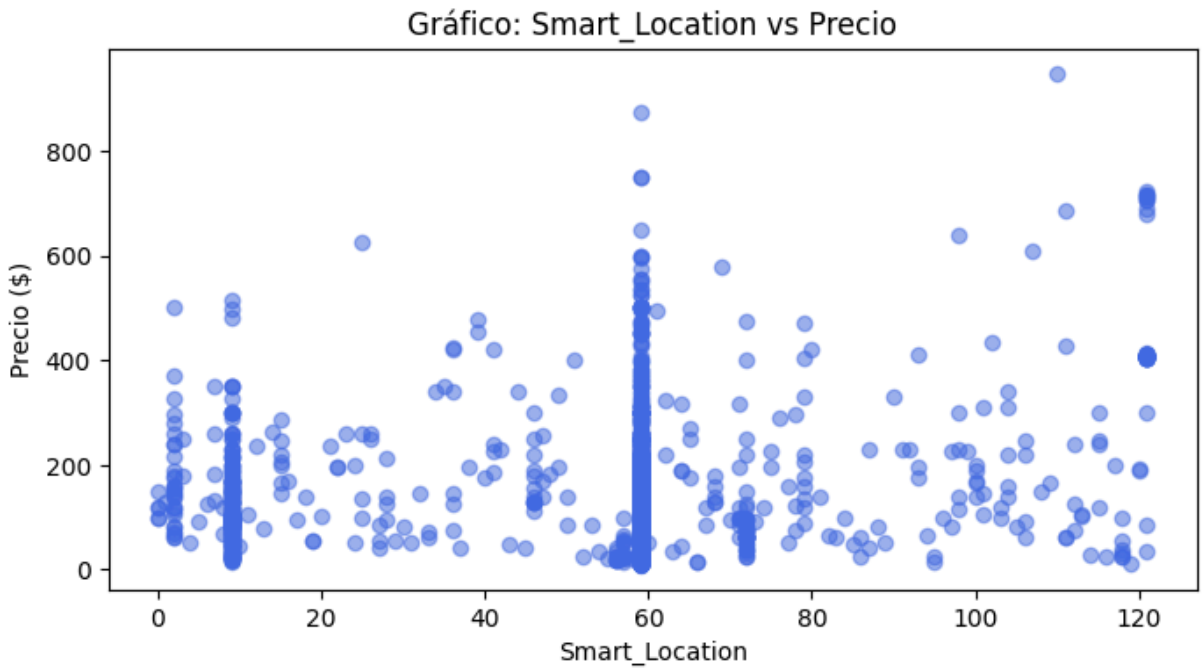


Gráfico: Smart_Location vs Precio
Precio ($)
0
200
400
600
800
0
20
40
60
80
100
120
Smart_Location

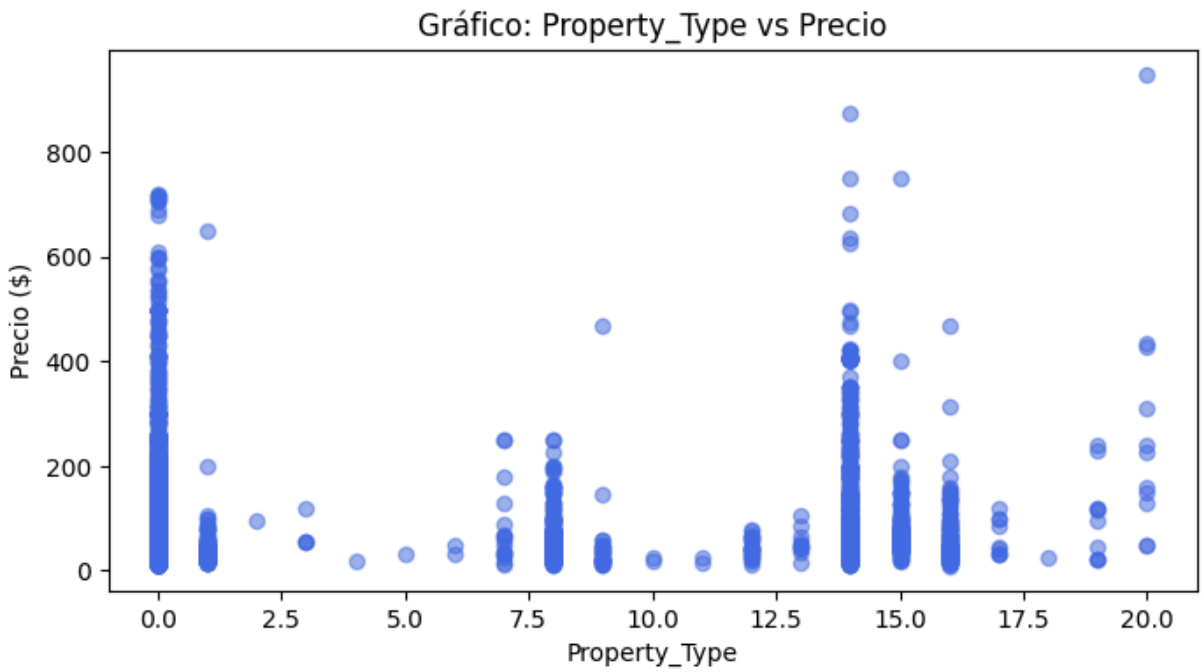


Gráfico: Property_Type vs Precio
Precio ($)
0
200
400
600
800
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Property_Type

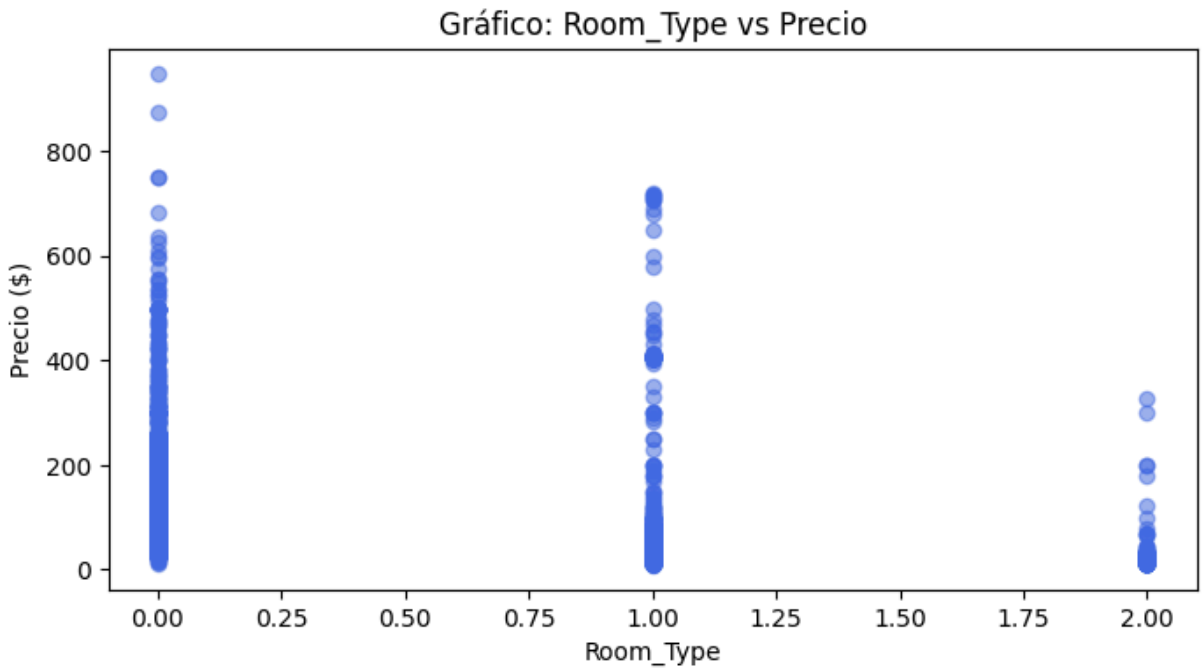


Gráfico: Room_Type vs Precio
Precio ($)
0
200
400
600
800
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Room_Type

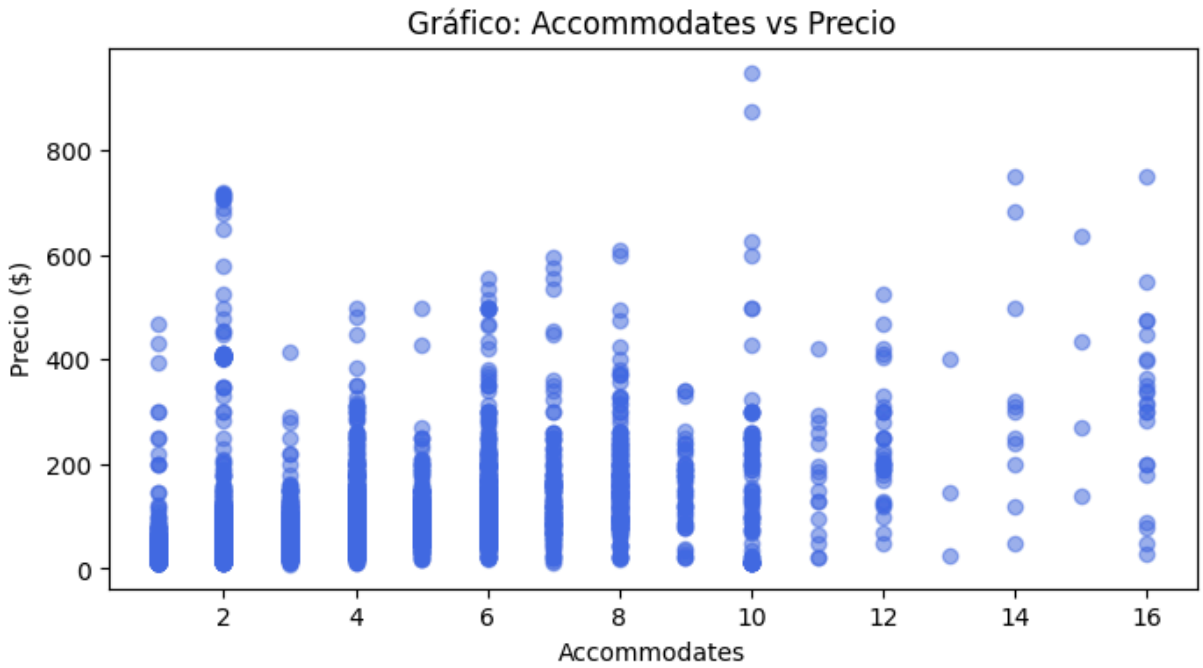


Gráfico: Accommodates vs Precio
Precio ($)
0
200
400
600
800
2
4
6
8
10
12
14
16
Accommodates

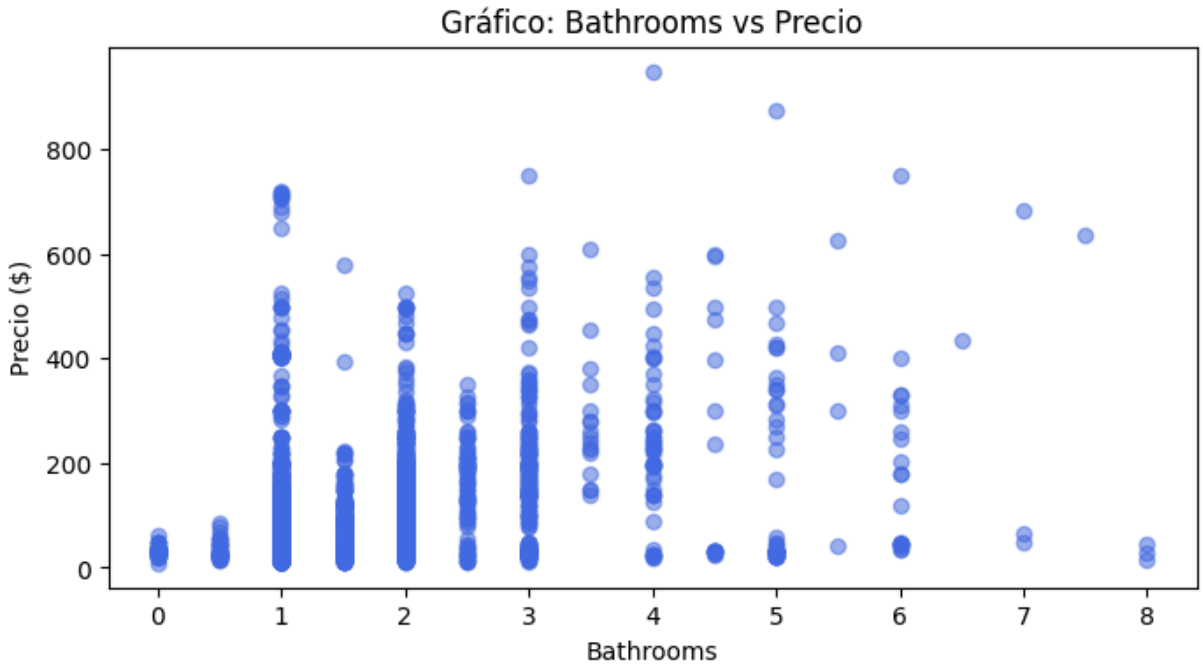


Gráfico: Bathrooms vs Precio
Precio ($)
0
200
400
600
800
0
1
2
3
4
5
6
7
8
Bathrooms

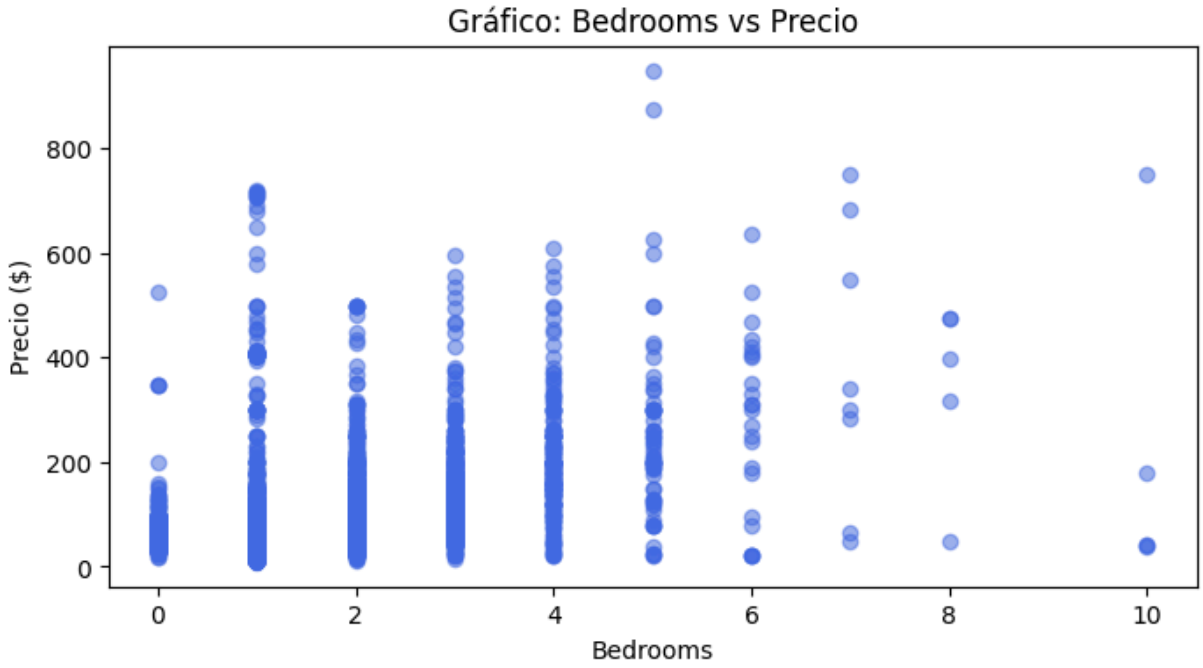
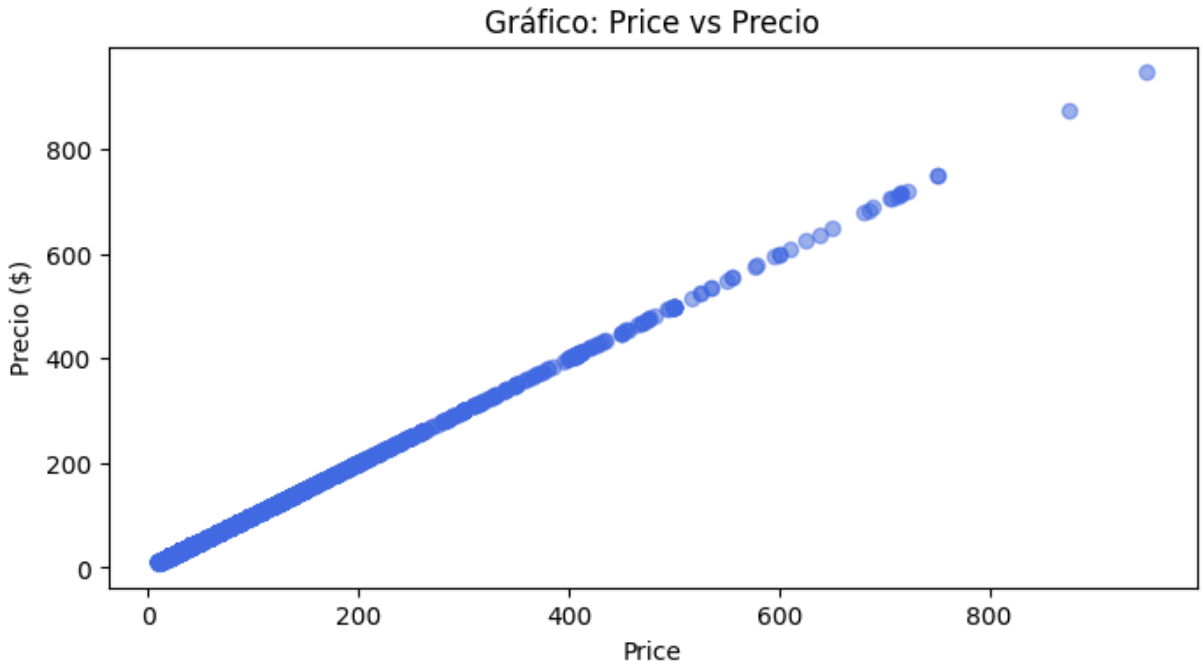
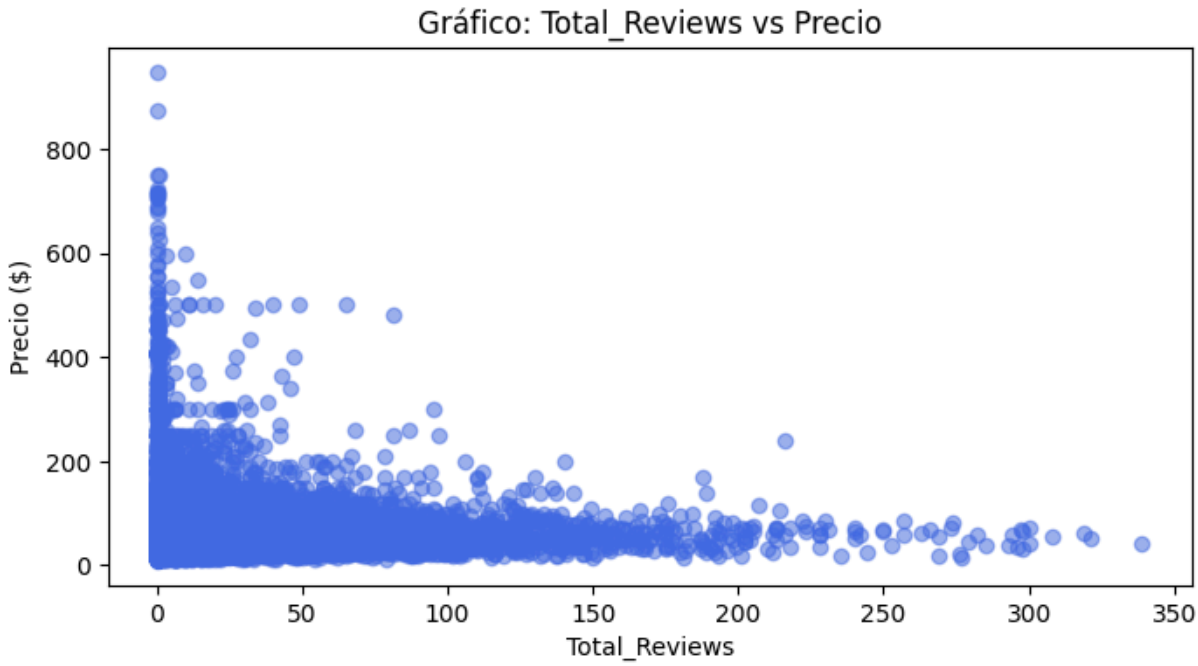
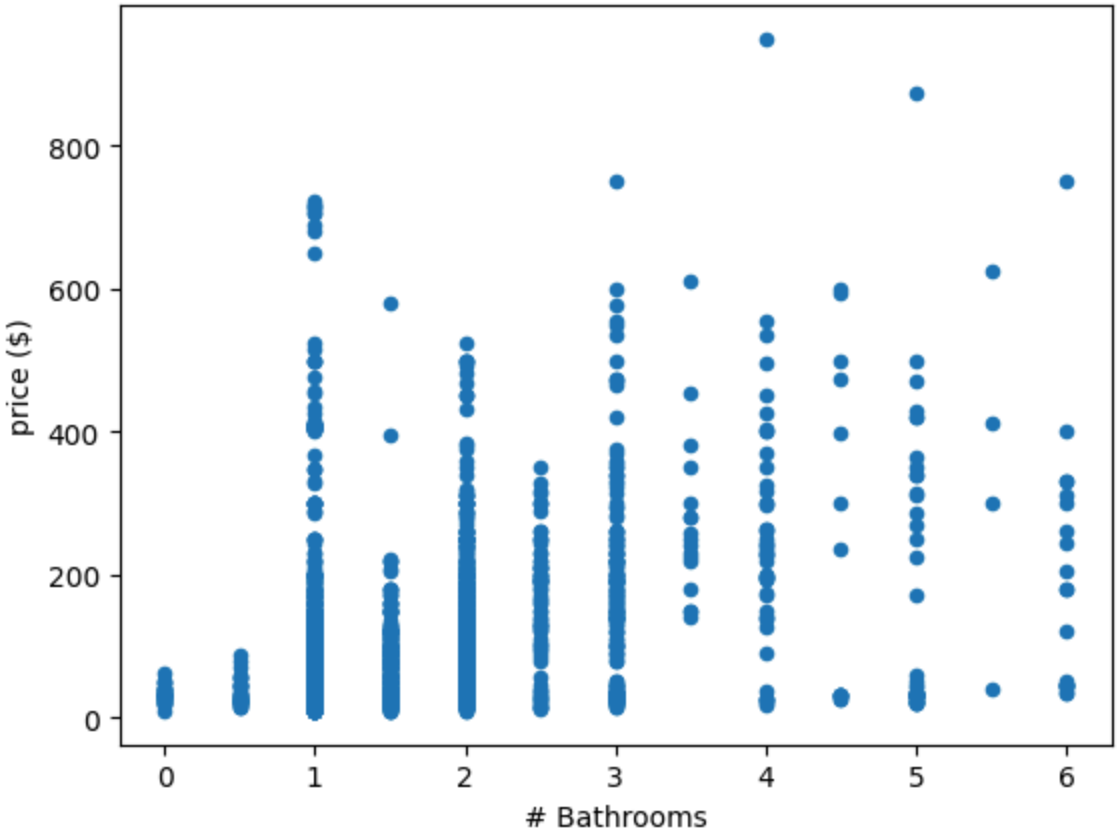


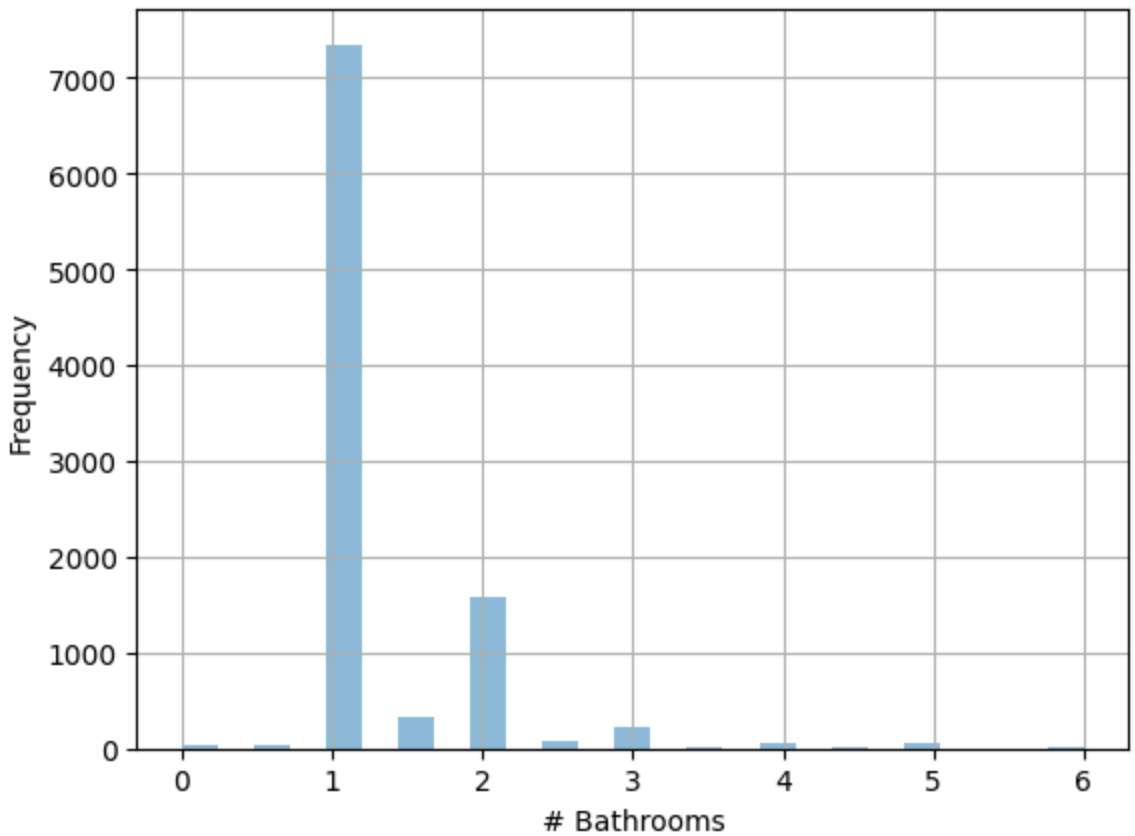
Gráfico: Bedrooms vs Precio
Precio ($)
0
200
400
600
800
0
2
4
6
8
10
Bedrooms





In [60]:
```
#Limpiar Outliers Bathrooms
airbnb_df_no_outliers_bathrooms = airbnb_df[airbnb_df['Bathrooms'] <= 6]

airbnb_df_no_outliers_bathrooms.plot(kind = 'scatter',x='Bathrooms',y = 'Price')
plt.xlabel('# Bathrooms')
plt.ylabel('price ($)')
plt.show()

airbnb_df_no_outliers_bathrooms['Bathrooms'].plot.hist(alpha=0.5, bins=25, grid = T
plt.xlabel('# Bathrooms')
plt.show()
```





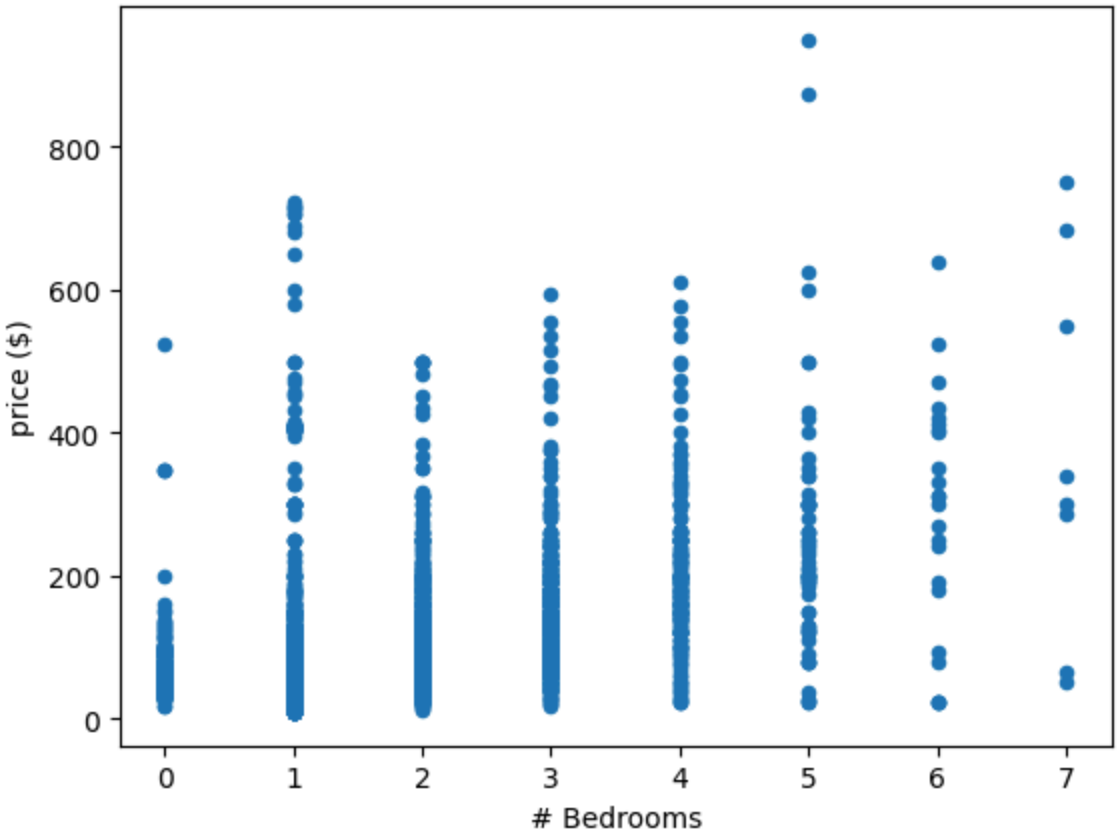
In [61]:
```
print(f'Outliers Bathrooms')
print(
    f'Original: {airbnb_df.shape[0]} // '
    f'Modificado: {airbnb_df_no_outliers_bathrooms.shape[0]}\nDiferencia: {airbnb_d
```

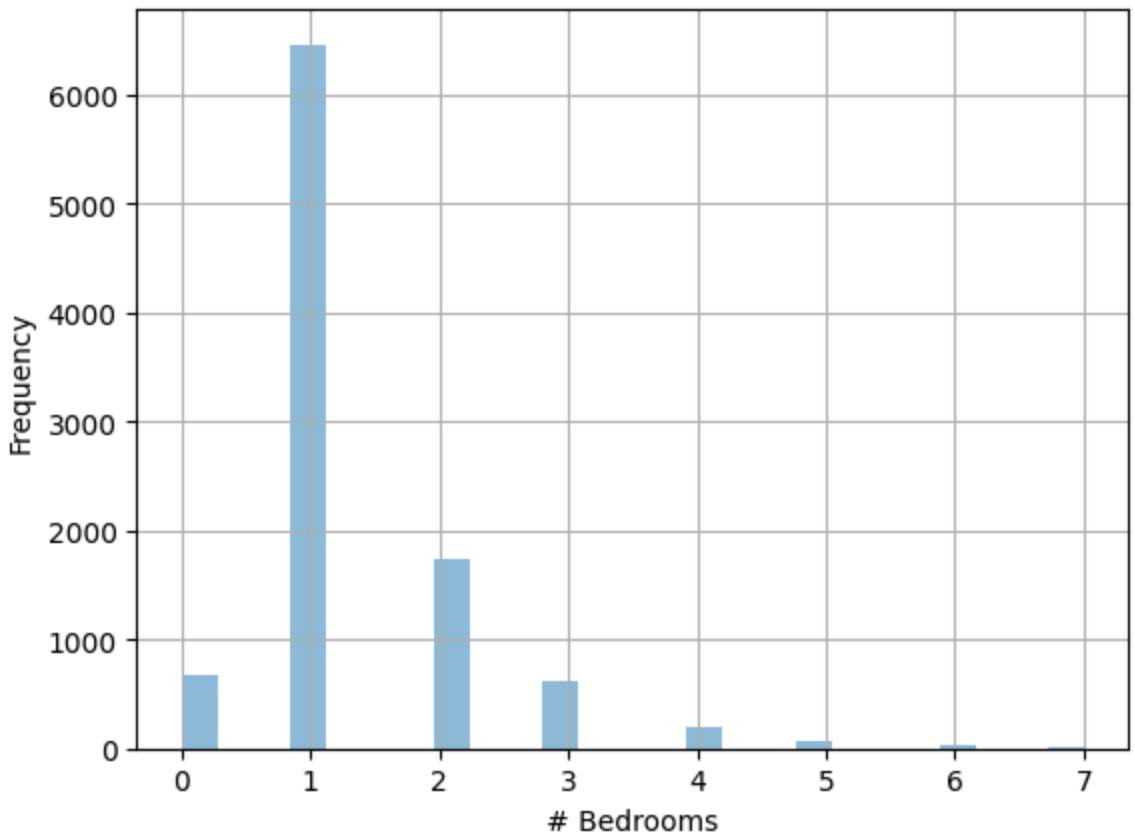
```
)
print(f'Variación: {((airbnb_df.shape[0] - airbnb_df_no_outliers_bathrooms.shape[0]
```

```
Outliers Bathrooms
Original: 9818 // Modificado: 9810
Diferencia: 8
Variación: 0.081483%
```

In [62]:

```
#Limpiar Outliers Bedrooms
airbnb_df_no_outliers_bedrooms = airbnb_df[airbnb_df['Bedrooms'] <= 7]

airbnb_df_no_outliers_bedrooms.plot(kind = 'scatter',x='Bedrooms',y = 'Price')
plt.xlabel('# Bedrooms')
plt.ylabel('price ($)')
plt.show()

airbnb_df_no_outliers_bedrooms['Bedrooms'].plot.hist(alpha=0.5, bins=25, grid = Tru
plt.xlabel('# Bedrooms')
plt.show()
```





In [63]:
```
print(f'Outliers Bedrooms')
print(
    f'Original: {airbnb_df.shape[0]} // '
    f'Modificado: {airbnb_df_no_outliers_bedrooms.shape[0]}\nDiferencia: {airbnb_df
)
print(f'Variación: {((airbnb_df.shape[0] - airbnb_df_no_outliers_bedrooms.shape[0])
```

```
Outliers Bedrooms
Original: 9818 // Modificado: 9808
Diferencia: 10
Variación: 0.101854%
```

In [64]:
```
### Se actualiza dataset
airbnb_df = airbnb_df[airbnb_df['Bathrooms'] <= 6]
airbnb_df = airbnb_df[airbnb_df['Bedrooms'] <= 7]
airbnb_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 9801 entries, 0 to 10345
Data columns (total 11 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Street          9801 non-null   float64
 1   Neighbourhood   9801 non-null   float64
 2   City            9801 non-null   float64
 3   Smart_Location  9801 non-null   float64
 4   Property_Type   9801 non-null   float64
 5   Room_Type       9801 non-null   float64
 6   Accommodates    9801 non-null   int64
 7   Bathrooms       9801 non-null   float64
 8   Bedrooms        9801 non-null   float64
 9   Price           9801 non-null   float64
 10  Total_Reviews   9801 non-null   int64
dtypes: float64(9), int64(2)
memory usage: 918.8 KB
```

In [65]:

```
#Matriz de correlación
airbnb_df.corr()
```

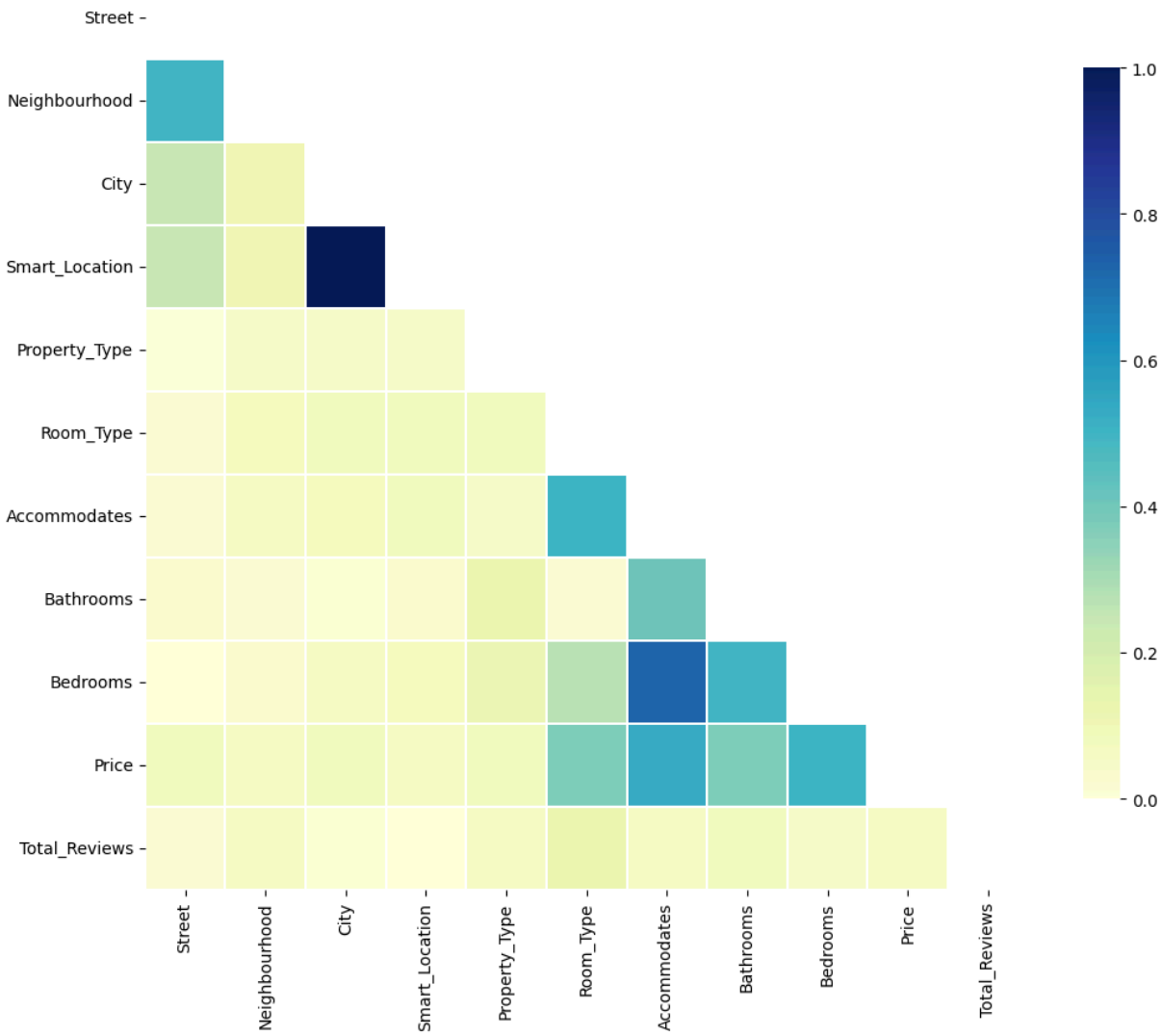
Out[65]:

	Street	Neighbourhood	City	Smart_Location	Property_Type	Roc
Street	1.000000	0.493833	0.245289	0.241742	-0.006667	-(
Neighbourhood	0.493833	1.000000	-0.101881	-0.104981	-0.049719	-(
City	0.245289	-0.101881	1.000000	0.997791	0.052964	(
Smart_Location	0.241742	-0.104981	0.997791	1.000000	0.040985	(
Property_Type	-0.006667	-0.049719	0.052964	0.040985	1.000000	(
Room_Type	-0.013544	-0.072087	0.081492	0.083905	0.080759	1
Accommodates	0.015553	0.064321	-0.071114	-0.082515	0.039678	-(
Bathrooms	0.034787	0.016934	-0.010966	-0.023687	0.128439	-(
Bedrooms	-0.001987	0.025174	-0.056457	-0.071121	0.110245	-(
Price	0.080689	0.057911	0.081943	0.055537	0.082259	-(
Total_Reviews	0.013142	0.060477	-0.007971	-0.002979	-0.063977	-(

In [66]:

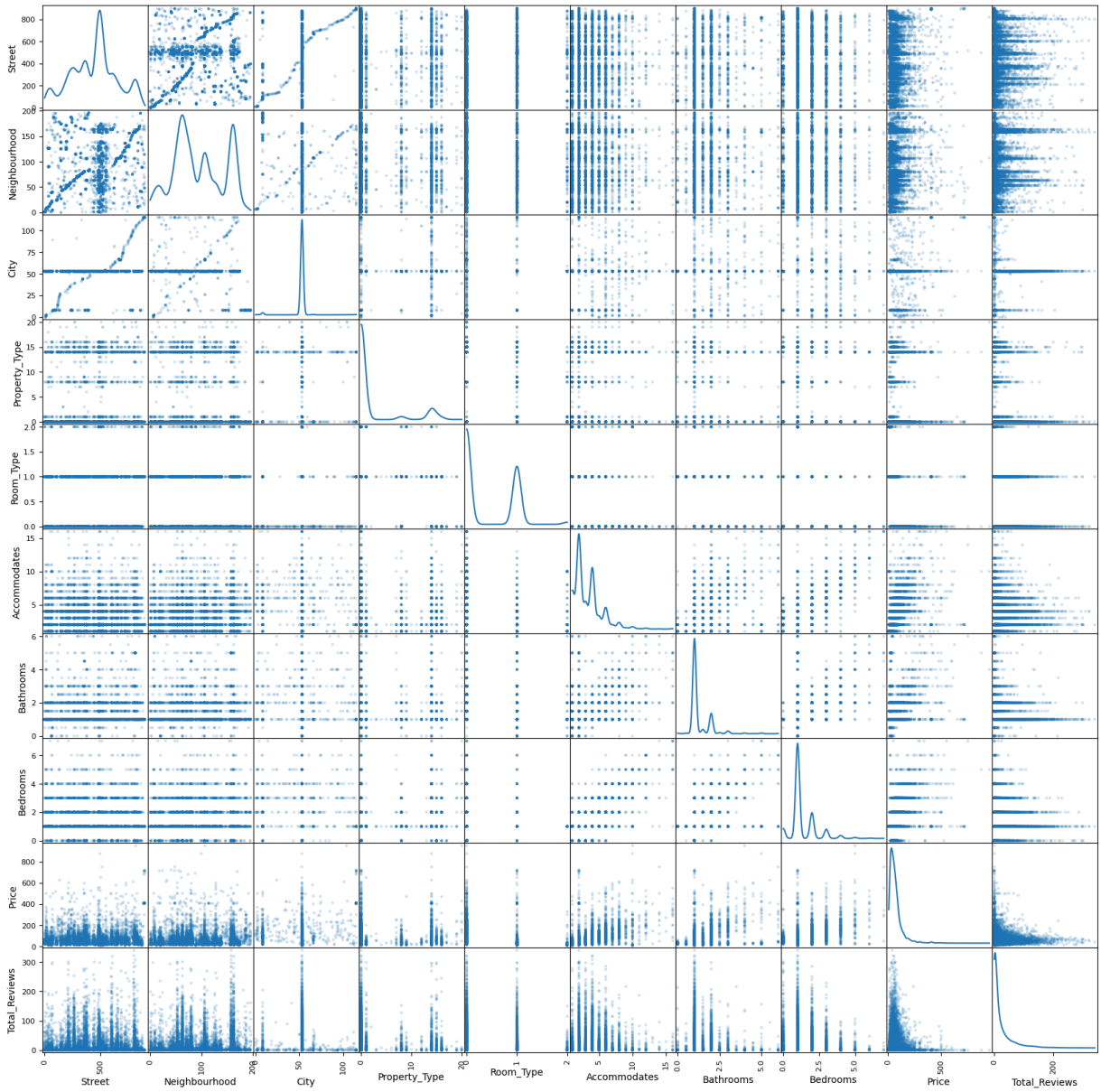
```
#Gráfica de correlación
import seaborn as sns

# Compute the correlation matrix
corr = np.abs(airbnb_df.corr())

# Generate a mask for the upper triangle
mask = np.zeros_like(corr, dtype=bool)
mask[np.triu_indices_from(mask)] = True

# Set up the matplotlib figure
```

```
f, ax = plt.subplots(figsize=(12, 10))

# Draw the heatmap with the mask and correct aspect ratio
sns.heatmap(corr, mask=mask,vmin = 0.0, vmax=1.0, center=0.5,
            linewidths=.1, cmap="YlGnBu", cbar_kws={"shrink": .8})

plt.show()
```



In [67]:
```
#Se elimina la columna que tienen la misma correlación por lo que seria redundante
airbnb_df = airbnb_df.drop(['Smart_Location'], axis=1)
#airbnb_df.info()
```

In [68]:
```
pd.plotting.scatter_matrix(airbnb_df, alpha=0.2, figsize=(20, 20), diagonal = 'kde'
plt.show()
```



4. Modelado, cross-validation y estudio de resultados en train y test

Proceso en el dataset Train

In [69]:
```
# Carga de datos
airbnb_data_full = pd.read_csv('./airbnb_train.csv', sep=';', decimal='.')

# Imputamos el pais y filtrmos
airbnb_data_full['Country'].fillna('Spain', inplace=True) # Omitir en test
airbnb_data_full = airbnb_data_full[airbnb_data_full['Country'] == 'Spain']

# Imputación
airbnb_data_full['City'].fillna('Madrid', inplace=True) # Omitir en test
airbnb_data_full['Bathrooms'].fillna(airbnb_data_full['Bathrooms'].mode()[0], inpla
airbnb_data_full['Bedrooms'].fillna(airbnb_data_full['Bedrooms'].mode()[0], inplace
airbnb_data_full['Price'].fillna(airbnb_data_full['Price'].mode()[0], inplace=True)
```

```
# Extraemos columnas que nos importan
airbnb_data = airbnb_data_full[['Price', 'Street', 'Neighbourhood Cleansed', 'City'
                               'Room Type','Accommodates', 'Bathrooms', 'Bedrooms',

# Renombramos columnas
airbnb_data.columns = ['Price', 'Street', 'Neighbourhood', 'City',  'Property_Type'
                       'Room_Type', 'Accommodates', 'Bathrooms', 'Bedrooms', 'Total

# Aplicamos el encoder
from sklearn.preprocessing import OrdinalEncoder
encoder = OrdinalEncoder()
categorical_cols = [
    'Street', 'Neighbourhood', 'City', 'Property_Type', 'Room_Type'
]
airbnb_data[categorical_cols] = encoder.fit_transform(airbnb_data[categorical_cols]

# Eliminamos outliers en bedrooms
airbnb_data = airbnb_data[airbnb_data['Bathrooms'] <= 6]
airbnb_data = airbnb_data[airbnb_data['Bedrooms'] <= 7]

# Verificamos el dataset
airbnb_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 9801 entries, 0 to 10345
Data columns (total 10 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Price          9801 non-null   float64
 1   Street         9801 non-null   float64
 2   Neighbourhood  9801 non-null   float64
 3   City           9801 non-null   float64
 4   Property_Type  9801 non-null   float64
 5   Room_Type      9801 non-null   float64
 6   Accommodates   9801 non-null   int64
 7   Bathrooms      9801 non-null   float64
 8   Bedrooms       9801 non-null   float64
 9   Total_Reviews  9801 non-null   int64
dtypes: float64(8), int64(2)
memory usage: 842.3 KB
```

Proceso en el dataset Test

In [70]:

```
# Carga de datos
airbnb_data_test_full = pd.read_csv('./airbnb_test.csv', sep=';', decimal='.')

# Imputamos el pais y filtrmos
airbnb_data_test_full = airbnb_data_test_full[airbnb_data_test_full['Country'] == '

# Imputación
airbnb_data_test_full['Bathrooms'].fillna(airbnb_data_test_full['Bathrooms'].mode()
airbnb_data_test_full['Bedrooms'].fillna(airbnb_data_test_full['Bedrooms'].mode()[0
airbnb_data_test_full['Price'].fillna(airbnb_data_test_full['Price'].mode()[0], inp
```

```
# Extraemos columnas que nos importan
airbnb_data_test = airbnb_data_test_full[['Price', 'Street', 'Neighbourhood Cleanse
                                          'Room Type','Accommodates', 'Bathrooms',

# Renombramos columnas
airbnb_data_test.columns = ['Price', 'Street', 'Neighbourhood', 'City',  'Property_
                            'Room_Type', 'Accommodates', 'Bathrooms', 'Bedrooms', '

# Aplicamos el encoder
airbnb_data_test[categorical_cols] = encoder.fit_transform(airbnb_data_test[categor

# Diligenciamos los null
airbnb_data_test['City'].fillna(airbnb_data_test['City'].mode()[0], inplace=True)

# Eliminamos outliers en bedrooms
airbnb_data_test = airbnb_data_test[airbnb_data_test['Bathrooms'] <= 6]
airbnb_data_test = airbnb_data_test[airbnb_data_test['Bedrooms'] <= 7]

# Verificamos el dataset
airbnb_data_test.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 4176 entries, 0 to 4433
Data columns (total 10 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Price          4176 non-null   float64
 1   Street         4176 non-null   float64
 2   Neighbourhood  4176 non-null   float64
 3   City           4176 non-null   float64
 4   Property_Type  4176 non-null   float64
 5   Room_Type      4176 non-null   float64
 6   Accommodates   4176 non-null   int64
 7   Bathrooms      4176 non-null   float64
 8   Bedrooms       4176 non-null   float64
 9   Total_Reviews  4176 non-null   int64
dtypes: float64(8), int64(2)
memory usage: 358.9 KB
```

In [71]:

```
# Dataset de train
data_train = airbnb_data.values
X_train_raw = data_train[:,1:]      # nos quedamos con el resto
y_train = data_train[:,0:1]     # nos quedamos con la 1ª columna, price

# Dataset de test
data_test = airbnb_data_test.values
X_test_raw = data_test[:,1:]      # nos quedamos con el resto
y_test = data_test[:,0:1]     # nos quedamos con la 1ª columna, price
```

In [72]:

```
from sklearn import preprocessing

# Escalamos los datos y se normalizan
scaler = preprocessing.StandardScaler().fit(X_train_raw)
X_train = scaler.transform(X_train_raw)

# recordad que esta normalización/escalado la realizo con el scaler anterior, basad
```

```
X_test = scaler.transform(X_test_raw)

# Visualización de los datos
print('Datos entrenamiento: ', X_train.shape)
print('Datos test: ', X_test.shape)
```

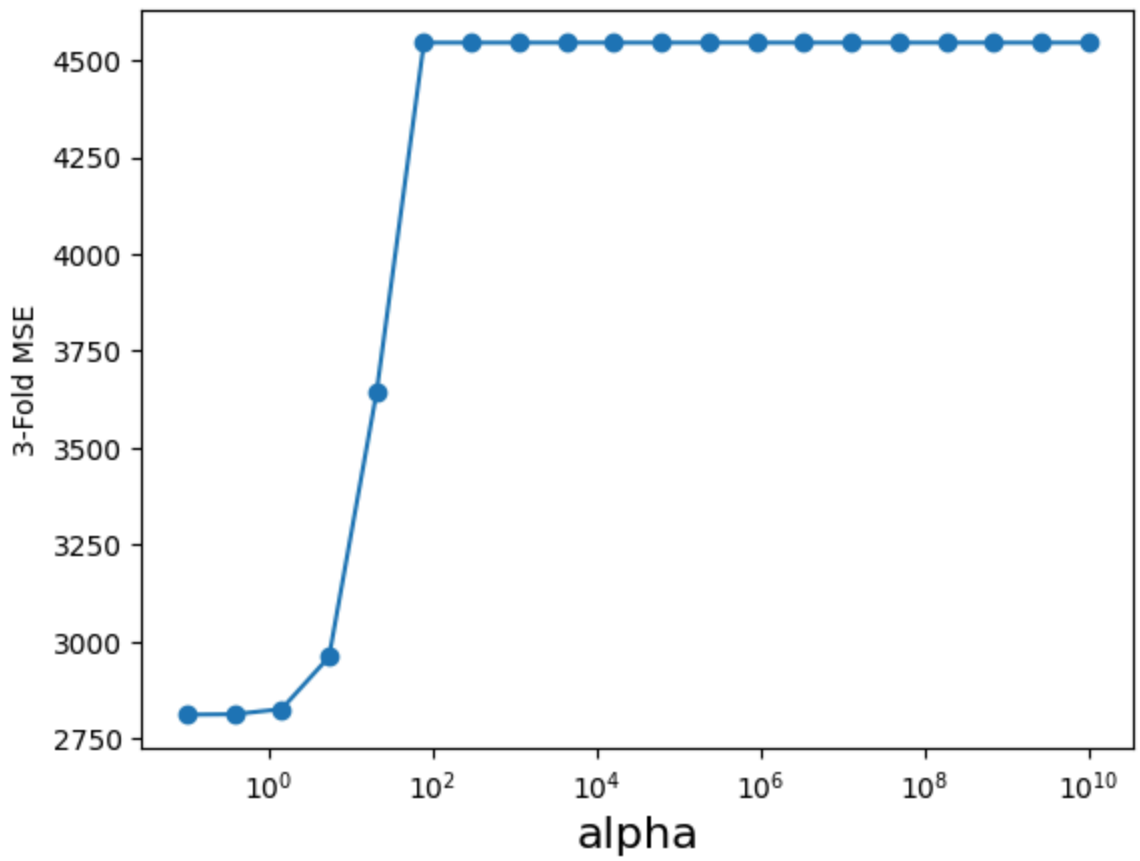
```
Datos entrenamiento:  (9801, 9)
Datos test:  (4176, 9)
```

4.1 Implementación de Lasso

In [73]:
```
from sklearn.model_selection import GridSearchCV
from sklearn.linear_model import Lasso

alpha_vector = np.logspace(-1,10,20)
param_grid = {'alpha': alpha_vector }
grid = GridSearchCV(Lasso(), scoring= 'neg_mean_squared_error', param_grid=param_gr
grid.fit(X_train, y_train)
print("best mean cross-validation score: {:.3f}".format(grid.best_score_))
print("best parameters: {}".format(grid.best_params_))

#-1 porque es negado
scores = -1*np.array(grid.cv_results_['mean_test_score'])
plt.semilogx(alpha_vector,scores,'-o')
plt.xlabel('alpha',fontsize=16)
plt.ylabel('3-Fold MSE')
plt.show()
```

```
Fitting 3 folds for each of 20 candidates, totalling 60 fits
[CV] END .........................................alpha=0.1; total time=   0.0s
[CV] END .........................................alpha=0.1; total time=   0.0s
[CV] END .........................................alpha=0.1; total time=   0.0s
[CV] END ........................alpha=0.37926901907322497; total time=   0.0s
[CV] END ........................alpha=0.37926901907322497; total time=   0.0s
[CV] END ........................alpha=0.37926901907322497; total time=   0.0s
[CV] END ............................alpha=1.438449888287663; total time=   0.0s
[CV] END ............................alpha=1.438449888287663; total time=   0.0s
[CV] END ............................alpha=1.438449888287663; total time=   0.0s
[CV] END ............................alpha=5.45559478116852; total time=   0.0s
[CV] END ............................alpha=5.45559478116852; total time=   0.0s
[CV] END ............................alpha=5.45559478116852; total time=   0.0s
[CV] END ............................alpha=20.6913808111479; total time=   0.0s
[CV] END ............................alpha=20.6913808111479; total time=   0.0s
[CV] END ............................alpha=20.6913808111479; total time=   0.0s
[CV] END ............................alpha=78.47599703514615; total time=   0.0s
[CV] END ............................alpha=78.47599703514615; total time=   0.0s
[CV] END ............................alpha=78.47599703514615; total time=   0.0s
[CV] END ............................alpha=297.63514416313194; total time=   0.0s
[CV] END ............................alpha=297.63514416313194; total time=   0.0s
[CV] END ............................alpha=297.63514416313194; total time=   0.0s
[CV] END ............................alpha=1128.8378916846884; total time=   0.0s
[CV] END ............................alpha=1128.8378916846884; total time=   0.0s
[CV] END ............................alpha=1128.8378916846884; total time=   0.0s
[CV] END ............................alpha=4281.332398719395; total time=   0.0s
[CV] END ............................alpha=4281.332398719395; total time=   0.0s
[CV] END ............................alpha=4281.332398719395; total time=   0.0s
[CV] END ............................alpha=16237.767391887242; total time=   0.0s
[CV] END ............................alpha=16237.767391887242; total time=   0.0s
[CV] END ............................alpha=16237.767391887242; total time=   0.0s
[CV] END ............................alpha=61584.82110660267; total time=   0.0s
[CV] END ............................alpha=61584.82110660267; total time=   0.0s
[CV] END ............................alpha=61584.82110660267; total time=   0.0s
[CV] END ............................alpha=233572.14690901214; total time=   0.0s
[CV] END ............................alpha=233572.14690901214; total time=   0.0s
[CV] END ............................alpha=233572.14690901214; total time=   0.0s
[CV] END ............................alpha=885866.7904100833; total time=   0.0s
[CV] END ............................alpha=885866.7904100833; total time=   0.0s
[CV] END ............................alpha=885866.7904100833; total time=   0.0s
[CV] END ............................alpha=3359818.286283788; total time=   0.0s
[CV] END ............................alpha=3359818.286283788; total time=   0.0s
[CV] END ............................alpha=3359818.286283788; total time=   0.0s
[CV] END ............................alpha=12742749.857031321; total time=   0.0s
[CV] END ............................alpha=12742749.857031321; total time=   0.0s
[CV] END ............................alpha=12742749.857031321; total time=   0.0s
[CV] END ............................alpha=48329302.38571752; total time=   0.0s
[CV] END ............................alpha=48329302.38571752; total time=   0.0s
[CV] END ............................alpha=48329302.38571752; total time=   0.0s
[CV] END ............................alpha=183298071.08324376; total time=   0.0s
[CV] END ............................alpha=183298071.08324376; total time=   0.0s
[CV] END ............................alpha=183298071.08324376; total time=   0.0s
[CV] END ............................alpha=695192796.177562; total time=   0.0s
[CV] END ............................alpha=695192796.177562; total time=   0.0s
[CV] END ............................alpha=695192796.177562; total time=   0.0s
[CV] END ............................alpha=2636650898.730366; total time=   0.0s
```

```
[CV] END ..........................alpha=2636650898.730366; total time=   0.0s
[CV] END ..........................alpha=2636650898.730366; total time=   0.0s
[CV] END ............................alpha=10000000000.0; total time=   0.0s
[CV] END ............................alpha=10000000000.0; total time=   0.0s
[CV] END ............................alpha=10000000000.0; total time=   0.0s
best mean cross-validation score: -2810.035
best parameters: {'alpha': np.float64(0.1)}
```



In [74]:

```
#
from sklearn.metrics import mean_squared_error

alpha_optimo = grid.best_params_['alpha']
lasso = Lasso(alpha = alpha_optimo).fit(X_train, y_train)

ytrainLasso = lasso.predict(X_train)
ytestLasso  = lasso.predict(X_test)
mseTrainModelLasso = mean_squared_error(y_train, ytrainLasso)
mseTestModelLasso = mean_squared_error(y_test, ytestLasso)

print('MSE Modelo Lasso (train): %0.3g' % mseTrainModelLasso)
print('MSE Modelo Lasso (test) : %0.3g' % mseTestModelLasso)

print('RMSE Modelo Lasso (train): %0.3g' % np.sqrt(mseTrainModelLasso))
print('RMSE Modelo Lasso (test) : %0.3g' % np.sqrt(mseTestModelLasso))

feature_names = airbnb_data.columns[1:] # es igual en train y en test

w = lasso.coef_
```

```
for f,wi in zip(feature_names,w):
    print(f,wi)
```

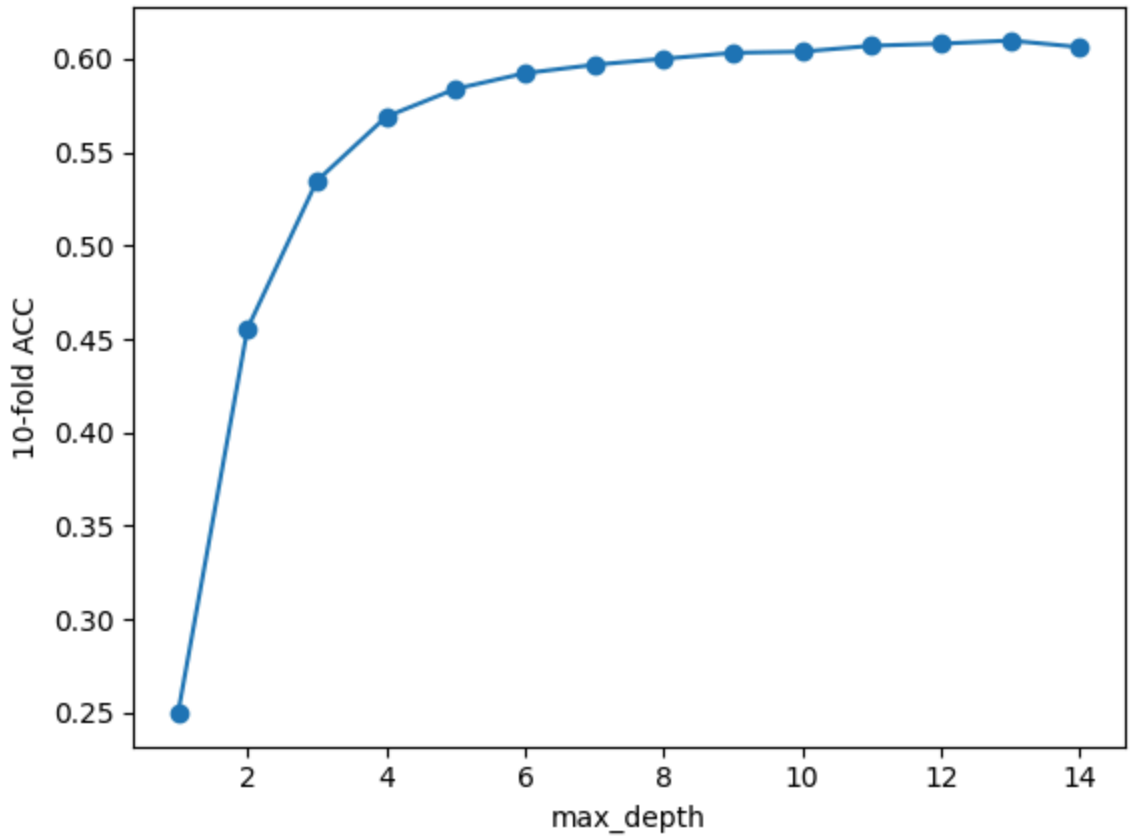
```
MSE Modelo Lasso (train): 2.79e+03
MSE Modelo Lasso (test) : 2.58e+03
RMSE Modelo Lasso (train): 52.8
RMSE Modelo Lasso (test) : 50.8
Street 1.8100284746698216
Neighbourhood 1.6247546478196315
City 8.125829387505599
Property_Type 2.498115435852801
Room_Type -15.753646532854443
Accommodates 14.308172369774065
Bathrooms 11.923824199847532
Bedrooms 12.862538030309475
Total_Reviews -5.69550773562406
```

4.2 Implementación de Random Forest

In [75]:

```
from sklearn.ensemble import RandomForestRegressor

maxDepth = range(1,15)
tuned_parameters = {'max_depth': maxDepth}

grid = GridSearchCV(RandomForestRegressor(random_state=0, n_estimators=200, max_fea
grid.fit(X_train, y_train)

print("best mean cross-validation score: {:.3f}".format(grid.best_score_))
print("best parameters: {}".format(grid.best_params_))

scores = np.array(grid.cv_results_['mean_test_score'])
plt.plot(maxDepth,scores,'-o')
plt.xlabel('max_depth')
plt.ylabel('10-fold ACC')

plt.show()
```

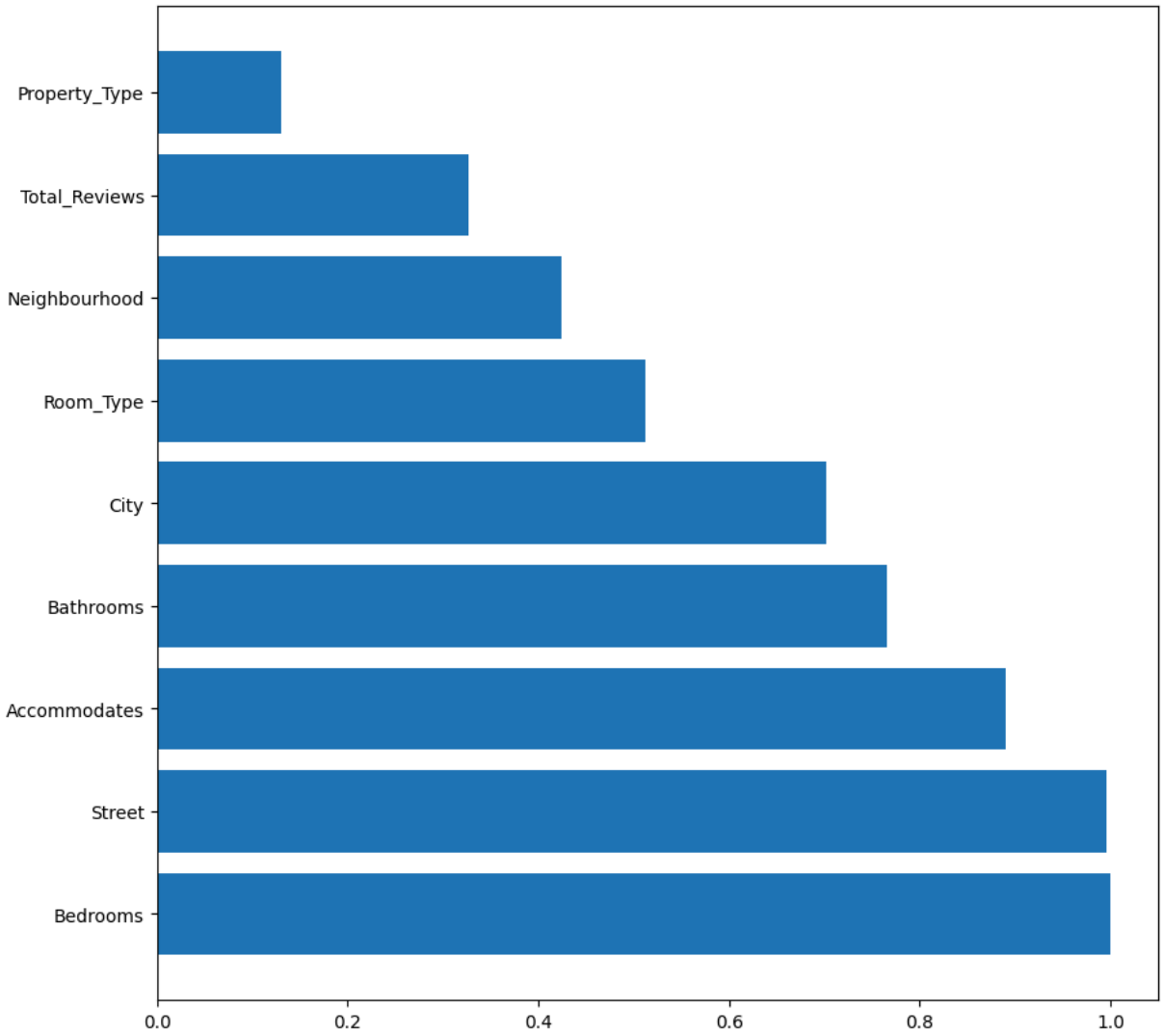
```
Fitting 3 folds for each of 14 candidates, totalling 42 fits
[CV] END .....................................max_depth=1; total time=   0.3s
[CV] END .....................................max_depth=1; total time=   0.2s
[CV] END .....................................max_depth=1; total time=   0.2s
[CV] END .....................................max_depth=2; total time=   0.3s
[CV] END .....................................max_depth=2; total time=   0.3s
[CV] END .....................................max_depth=2; total time=   0.3s
[CV] END .....................................max_depth=3; total time=   0.4s
[CV] END .....................................max_depth=3; total time=   0.4s
[CV] END .....................................max_depth=3; total time=   0.4s
[CV] END .....................................max_depth=4; total time=   0.4s
[CV] END .....................................max_depth=4; total time=   0.4s
[CV] END .....................................max_depth=4; total time=   0.4s
[CV] END .....................................max_depth=5; total time=   0.5s
[CV] END .....................................max_depth=5; total time=   0.5s
[CV] END .....................................max_depth=5; total time=   0.5s
[CV] END .....................................max_depth=6; total time=   0.6s
[CV] END .....................................max_depth=6; total time=   0.6s
[CV] END .....................................max_depth=6; total time=   0.6s
[CV] END .....................................max_depth=7; total time=   0.6s
[CV] END .....................................max_depth=7; total time=   0.6s
[CV] END .....................................max_depth=7; total time=  -0.5s
[CV] END .....................................max_depth=8; total time=   0.7s
[CV] END .....................................max_depth=8; total time=   0.7s
[CV] END .....................................max_depth=8; total time=   0.7s
[CV] END .....................................max_depth=9; total time=   0.8s
[CV] END .....................................max_depth=9; total time=   0.8s
[CV] END .....................................max_depth=9; total time=   0.8s
[CV] END ....................................max_depth=10; total time=   0.9s
[CV] END ....................................max_depth=10; total time=   0.9s
[CV] END ....................................max_depth=10; total time=   0.8s
[CV] END ....................................max_depth=11; total time=   0.9s
[CV] END ....................................max_depth=11; total time=   0.9s
[CV] END ....................................max_depth=11; total time=   0.9s
[CV] END ....................................max_depth=12; total time=   1.0s
[CV] END ....................................max_depth=12; total time=   1.0s
[CV] END ....................................max_depth=12; total time=   1.0s
[CV] END ....................................max_depth=13; total time=   1.1s
[CV] END ....................................max_depth=13; total time=   1.1s
[CV] END ....................................max_depth=13; total time=   1.1s
[CV] END ....................................max_depth=14; total time=   1.2s
[CV] END ....................................max_depth=14; total time=   1.2s
[CV] END ....................................max_depth=14; total time=   1.2s
best mean cross-validation score: 0.610
best parameters: {'max_depth': 13}
```



In [76]:
```
maxDepthOptimo = grid.best_params_['max_depth']
randomForest = RandomForestRegressor(max_depth=maxDepthOptimo,n_estimators=200,max_

print("Train: ",randomForest.score(X_train,y_train))
print("Test: ",randomForest.score(X_test,y_test))
```

```
Train:  0.8362642498803756
Test:  0.4945679831773465
```

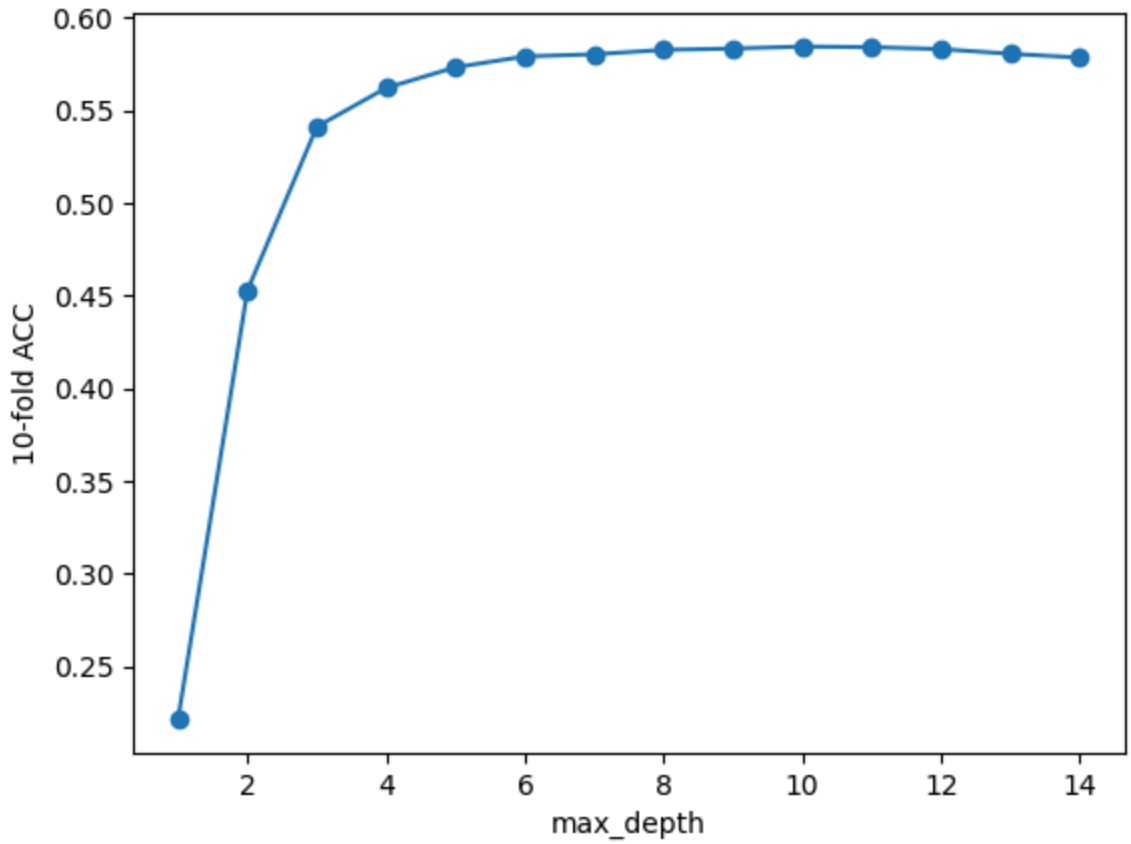
In [77]:
```
importances = randomForest.feature_importances_
importances = importances / np.max(importances)

indices = np.argsort(importances)[::-1]

plt.figure(figsize=(10,10))
plt.barh(range(X_train.shape[1]),importances[indices])
plt.yticks(range(X_train.shape[1]),feature_names[indices])
plt.show()
```



4.3 Implementación de Baggin regressor

In [78]:
```
from sklearn.ensemble import BaggingRegressor
from sklearn.model_selection import GridSearchCV
from sklearn.tree import DecisionTreeRegressor

# ... código aquí: GridSearch
maxDepth = range(1,15)
tuned_parameters = {'estimator__max_depth': maxDepth}

grid = GridSearchCV(BaggingRegressor(estimator=DecisionTreeRegressor(), random_stat
grid.fit(X_train, y_train)

print("best mean cross-validation score: {:.3f}".format(grid.best_score_))
print("best parameters: {}".format(grid.best_params_))

scores = np.array(grid.cv_results_['mean_test_score'])
plt.plot(maxDepth,scores,'-o')
plt.xlabel('max_depth')
plt.ylabel('10-fold ACC')

plt.show()
```

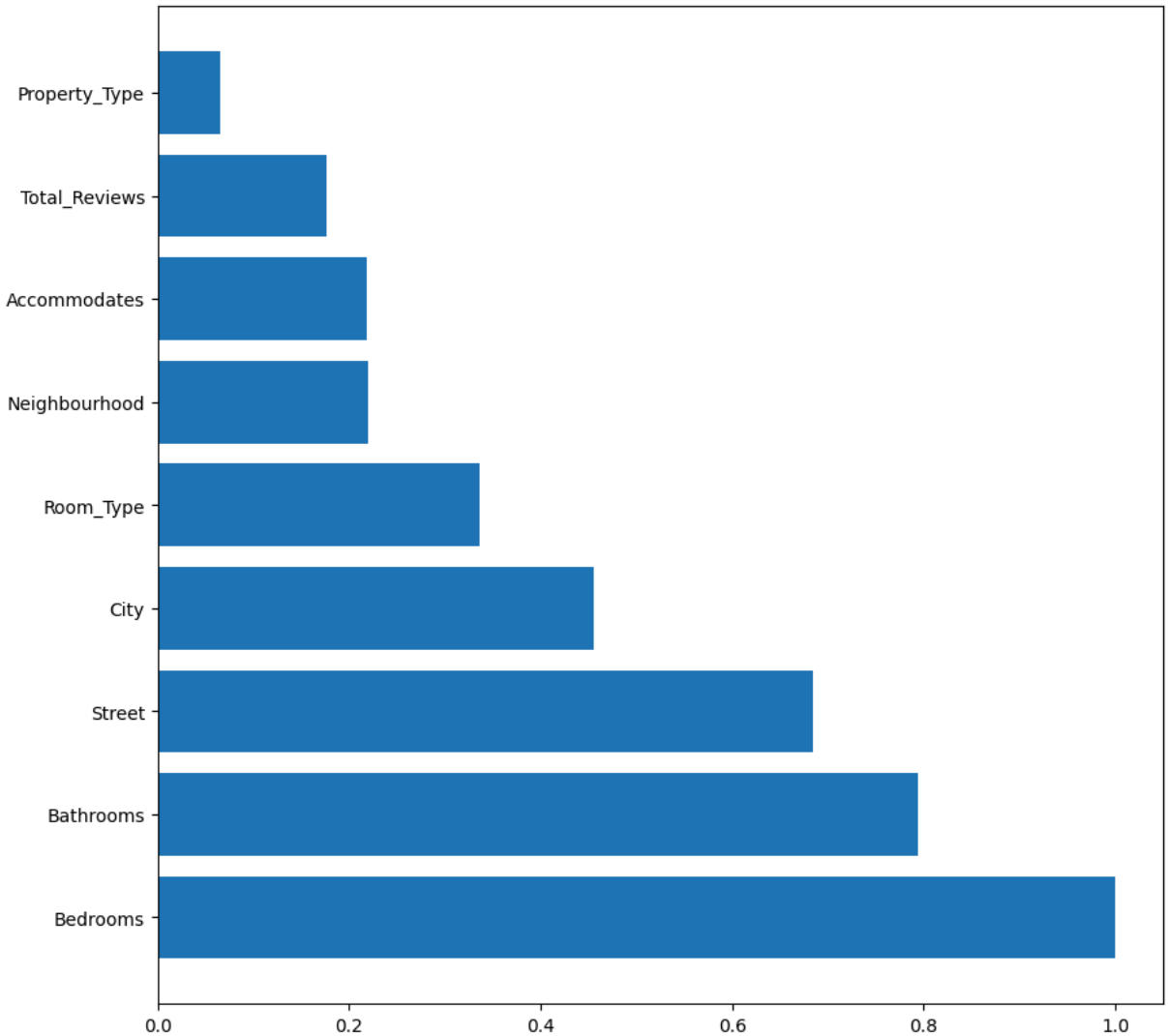
```
Fitting 3 folds for each of 14 candidates, totalling 42 fits
[CV] END ..............................estimator__max_depth=1; total time=   0.5s
[CV] END ..............................estimator__max_depth=1; total time=   0.5s
[CV] END ..............................estimator__max_depth=1; total time=   0.5s
[CV] END ..............................estimator__max_depth=2; total time=   0.6s
[CV] END ..............................estimator__max_depth=2; total time=   0.6s
[CV] END ..............................estimator__max_depth=2; total time=   0.6s
[CV] END ..............................estimator__max_depth=3; total time=   0.8s
[CV] END ..............................estimator__max_depth=3; total time=   0.8s
[CV] END ..............................estimator__max_depth=3; total time=   0.8s
[CV] END ..............................estimator__max_depth=4; total time=   0.9s
[CV] END ..............................estimator__max_depth=4; total time=   1.0s
[CV] END ..............................estimator__max_depth=4; total time=   0.9s
[CV] END ..............................estimator__max_depth=5; total time=   1.1s
[CV] END ..............................estimator__max_depth=5; total time=   1.1s
[CV] END ..............................estimator__max_depth=5; total time=   1.1s
[CV] END ..............................estimator__max_depth=6; total time=   1.3s
[CV] END ..............................estimator__max_depth=6; total time=   1.3s
[CV] END ..............................estimator__max_depth=6; total time=   1.3s
[CV] END ..............................estimator__max_depth=7; total time=   1.5s
[CV] END ..............................estimator__max_depth=7; total time=   1.5s
[CV] END ..............................estimator__max_depth=7; total time=   1.5s
[CV] END ..............................estimator__max_depth=8; total time=   1.7s
[CV] END ..............................estimator__max_depth=8; total time=   1.7s
[CV] END ..............................estimator__max_depth=8; total time=   1.7s
[CV] END ..............................estimator__max_depth=9; total time=   1.8s
[CV] END ..............................estimator__max_depth=9; total time=   1.8s
[CV] END ..............................estimator__max_depth=9; total time=   1.8s
[CV] END ............................estimator__max_depth=10; total time=   0.9s
[CV] END ............................estimator__max_depth=10; total time=   2.0s
[CV] END ............................estimator__max_depth=10; total time=   2.1s
[CV] END ............................estimator__max_depth=11; total time=   2.2s
[CV] END ............................estimator__max_depth=11; total time=   2.2s
[CV] END ............................estimator__max_depth=11; total time=   2.2s
[CV] END ............................estimator__max_depth=12; total time=   2.5s
[CV] END ............................estimator__max_depth=12; total time=   2.4s
[CV] END ............................estimator__max_depth=12; total time=   2.5s
[CV] END ............................estimator__max_depth=13; total time=   2.6s
[CV] END ............................estimator__max_depth=13; total time=   2.6s
[CV] END ............................estimator__max_depth=13; total time=   2.6s
[CV] END ............................estimator__max_depth=14; total time=   2.8s
[CV] END ............................estimator__max_depth=14; total time=   1.7s
[CV] END ............................estimator__max_depth=14; total time=   2.8s
best mean cross-validation score: 0.584
best parameters: {'estimator__max_depth': 10}
```



In [79]:
```
maxDepthOptimo = grid.best_params_['estimator__max_depth']
baggingRegressor = BaggingRegressor(estimator=DecisionTreeRegressor(max_depth=maxDe

print("Train: ", baggingRegressor.score(X_train, y_train))
print("Test: ", baggingRegressor.score(X_test, y_test))
```

```
Train:  0.7846359195100014
Test:  0.5060153056287833
```

In [80]:
```
importances = np.mean([tree.feature_importances_ for tree in baggingRegressor.estim
importances = importances / np.max(importances)

indices = np.argsort(importances)[::-1]

plt.figure(figsize=(10,10))
plt.barh(range(X_train.shape[1]),importances[indices])
plt.yticks(range(X_train.shape[1]),feature_names[indices])
plt.show()
```


Property_Type
Total_Reviews
Accommodates
Neighbourhood
Room_Type
City
Street
Bathrooms
Bedrooms
0.0
0.2
0.4
0.6
0.8
1.0

In [ ]: